

Learning Simple Test-Time Environments for LLM Web Agents

Junxuan Li^{1*†} Zijun Liu^{2*} Ziyi Huang^{3†} Peng Li⁴ Yuzhou Liu¹ Ming Yan⁵ Yang Liu^{2,4}

¹College of Computer Science and Technology, Jilin University, Jilin, China

²Dept. of Comp. Sci. & Tech., Tsinghua University, Beijing, China

³School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing, China

⁴Institute for AI Industry Research (AIR), Tsinghua University, Beijing, China

⁵Tongyi Lab, Alibaba Group

*Equal contribution. †Work done at AIR, Tsinghua University.

Abstract

Large language model (LLM) agents have demonstrated remarkable proficiency in manually constructed environments, yet their performance frequently collapses when transitioned to complex real-world settings. Existing research largely attribute this degradation to the compositional generalization gaps in LLMs on combinations of multiple simple, well-structured environments. In this work, we propose that LLM web agents can learn simple environment observations at test time. Specifically, we introduce trial steps for agents to decompose a complex environment observation into sub-modules, and implement a label-free learning method, Test-Time Environment Decomposition (**TTED**), to adapt agent behaviors with experience during inference. Our empirical evaluations demonstrate the framework’s efficacy across both synthetic and realistic benchmarks, showing (1) experience gains acquired within simpler sub-environments can be effectively composed to improve performance in the full one, and (2) test-time training on sub-environments can significantly enhance the compositional generalization of agents in real-world web automation tasks. We also provide key insights in the design of the label-free learning algorithm. As more complex environments are accessed by LLM agents, we believe learning environment decomposition skills at test time will be critical for robust real-world deployment.¹

Introduction

Large language model (LLM) agents have recently shown strong capabilities in sequential decision making (Xi et al. 2025; OpenAI 2026a), especially in environments where the state space is manually curated, interface layouts are regularized, and task structure is relatively local (Shi et al. 2017; Deng et al. 2023; Towers et al. 2025; Liu et al. 2025b). Such progress has led to growing optimism that language agents can serve as a general interface for web automation (He et al. 2024; Wei et al. 2025) and other tasks in diverse interactive environments (Wang et al. 2025; He et al. 2025).

However, a substantial gap remains between performance in these controlled settings and performance in realistic environments (Zhou et al. 2024; Drouin et al. 2024). A central reason for this failure is that real-world environments are

compositionally more complex than synthetic ones. In realistic web websites, a single observation may simultaneously contain multiple functional sub-structures, including search panels, form fields, content containers, modal windows, and other task-specific widgets. While each sub-structure may be easy to handle in isolation, their joint presence challenges the compositional generalization ability of current LLM agents. As a result, an agent that succeeds in individual sub-environments may still collapse in a cluttered interface (Furuta et al. 2024; Boisvert et al. 2024).

Through large-scale post-training (Shen et al. 2025; OpenAI 2026a), the base policy can be substantially improved, but further scaling is ultimately constrained by the limited availability of reliable ground-truth reward signals. Existing approaches seek to address this problem through more sophisticated test-time learning strategies (Pan et al. 2024; Yang et al. 2025b; Erdogan et al. 2025; Chae et al. 2025; He et al. 2025; Lee et al. 2026), including prompting, sampling, planning, etc. These methods improve task performance during inference, without external reward feedback. However, the core compositional generalization gap remains. As shown in the example in Figure 1 (a), (1) the agent’s exploration success rate degrades when the environment is compositionally complex, resulting in bottlenecked exploration capabilities; and (2) the agent’s self-assessment (Zheng et al. 2023) accuracy also degrades, co-leading to ineffective test-time learning. These limitations become particularly severe in web automation, where relevant decision variables are often localized, while irrelevant context is abundant. However, as an intuitive but unexplored solution, it remains unclear **whether an agent can learn to simplify environment observations at test time and thus mitigate these failures.**

In this work, we propose **Test-Time Environment Decomposition (TTED)**, a label-free learning framework that adapts agent behavior within decomposed sub-environments during inference. Instead of asking the agent to directly solve the full environment in one pass, we introduce an explicit trial step that the agent partitions a complex observation into a set of specialized and manageable sub-modules. These sub-modules serve as local decision contexts in which the agent can refine its behavior before returning to the full task. Specifically, TTED adapts the agent policy using self-assessment feedback derived from interaction experience within the sub-environments, either by in-context

¹Code and data are released at <https://github.com/THUNLP-MT/TTED>.

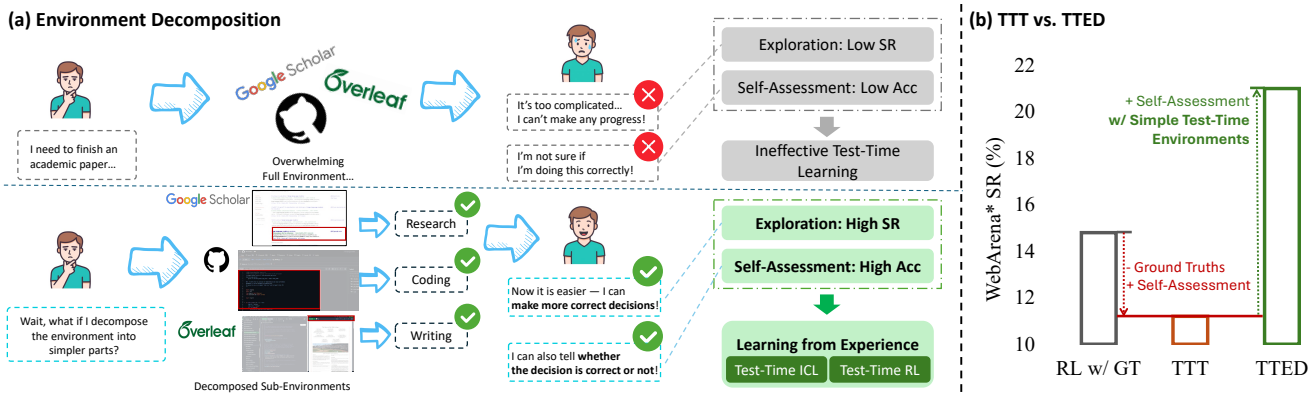


Figure 1: An overview of (a) environment decomposition and a performance comparison of (b) test-time training (TTT) and TTED. “SR” denotes success rate. “GT” denotes ground truth. (a) In complex real-world tasks, e.g., writing academic papers, the environment is often compositionally complex. Decomposing into simple test-time environments can help the agent explore and self-assess more effectively, resulting in improved performance after test-time learning. (b) TTED recovers performance degradation of TTT and outperforms RL on full ground-truth rewards. *We only use a subset of WebArena for evaluation.

learning (ICL) (Brown et al. 2020; Dong et al. 2024), which works for single-turn, static webpages, or by reinforcement learning (RL), which collects experience across multiple steps and tasks to fit multi-turn realistic web environments.

We evaluate TTED on both synthetic and realistic web benchmarks. We extend task quantity in CompWoB (Furuta et al. 2024) with an automated synthesis engine to feature more complex environment compositions, resulting in *CompWoB+*. Empirically, we show that TTED with ICL can substantially alleviate the performance degradation when environment complexity increases, showing the capacity of LLMs to perform effective decomposition. We further show that TTED with RL can significantly improve generalization on realistic web automation tasks on WebArena (Zhou et al. 2024), even outperforming RL trained with ground-truth rewards (Figure 1 (b)). We also provide detailed analysis on LLM self-assessment quality, out-of-distribution generalization on WorkArena (Drouin et al. 2024), data and environment scaling effects, and the design of RL algorithms.

In summary, our key contributions are as follows:

- We introduce TTED as a test-time learning solution to compositional generalization failures in LLM web agents.
- We extend a larger synthetic benchmark, *CompWoB+*, to systematically evaluate the impact of environment composition on agent performance and show that TTED can effectively mitigate the resulting generalization gap.
- We demonstrate TTED with proper RL design is effective in realistic web automation tasks such as WebArena, even outperforming RL trained with ground-truth rewards.
- We analyze several key factors that drive the success of test-time adaptation under environment decomposition.

Related Work

Environment Generalization in LLM Web Agents The transition from synthetic to real-world environments exposes a critical vulnerability in LLM agents: environment generalization. LLMs can recombine learned primitives in con-

trolled settings (Wei et al. 2022; Abedsoltan et al. 2025), and multi-agent RL can improve transfer across distinct environments (He et al. 2025). However, recent work (Boisvert et al. 2024; Xu et al. 2025) suggests that LLMs often exhibit brittle generalization when tasks require systematic composition across multiple web modules, pages, or websites. This challenge is acute in realistic websites, where task-relevant modules are buried in irrelevant context; even strong proprietary models remain weak on end-to-end knowledge-work automation (Boisvert et al. 2024). Maintaining both reliable action generation and accurate self-assessment in such cluttered environments therefore remains an open challenge.

Test-Time Learning for LLM Web Agents Test-time learning adapts agent behavior during inference using signals available from test instances rather than ground-truth labels (Akyurek et al. 2025; Yuksekgonul et al. 2026). For web agents, these methods include prompting, sampling, planning, and reflection (Pan et al. 2024; Erdogan et al. 2025; Chae et al. 2025; Lee et al. 2026). ICL (Brown et al. 2020) methods optimize the agent’s behavior at test time on each query individually, which can be termed as *query-level* test-time learning. However, they are constrained to benefit from experience within multiple different queries. Test-time training (Akyurek et al. 2025; Zuo et al. 2025) instead updates model parameters from experience accumulated across queries. However, existing approaches either assume synthetic task formats (Akyurek et al. 2025) or rely on matching-based self-consistency rewards (Zuo et al. 2025), which cannot directly supervise free-form environment decomposition. Although test-time training can improve generalization under distribution shift (Sun et al. 2020), its effectiveness depends on accurate and informative learning signals, which are often sparse and noisy in complex environments.

Positioning of Our Work TTED connects these directions by decomposing a complex environment into an aligned sub-goal and sub-observation, collecting self-assessed experience in this local context, and adapting the agent through

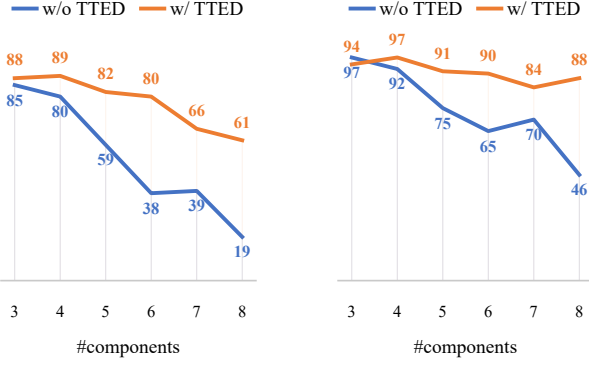


Figure 2: Task success rates (%) on CompWoB+ with Qwen3-8B (left) and GPT-4o-mini (right). TTED effectively mitigates compositional generalization failures on both models.

query-level ICL or task-level RL. Unlike planning and reflection (Erdogan et al. 2025; Pan et al. 2024) typically modify reasoning, action search, or feedback while retaining the full environment representation, TTED decomposes the representation for both action generation and self-assessment. It also differs from history shrinking or context pruning (Yang et al. 2025b): localized sub-environments are not merely shorter inputs but test-time learning units. TTED therefore directly targets the coupled degradation of exploration and self-assessment under compositional complexity. Further distinctions from planning are discussed in Appendix A.1.

Preliminaries

Test-Time Environment

We model the test-time environment of a long-horizon web task as a Markov decision process (MDP) $\mathcal{E} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R})$. A test-time environment refers to the task-conditioned decision process encountered by the policy during inference, including the interaction context observable to the agent, the available actions, the environment transitions, and the reward signals associated with the task. At step t , the state is

$$s_t = (g, o_t, H_{t-1}), \quad (1)$$

where g , o_t , and H_{t-1} denote the task goal, current observation, and interaction history, respectively. A policy $\pi_\theta(a_t | s_t)$ induces a trajectory $\tau = (o_1, a_1, \dots, o_T, a_T)$. A conventional approach directly optimizes the policy in the original test-time environment:

$$\pi_{\text{ori}}^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} [\mathcal{R}(\tau)]. \quad (2)$$

However, abundant irrelevant information and multi-step execution in complex web environments make task rewards sparse and delayed, while also complicating accurate step-level reward estimation. Consequently, direct policy optimization in the full test-time environment is difficult.

To address this issue, TTED decomposes the original test-time environment into multiple aligned local decision con-

texts, each consisting of a sub-goal and its task-relevant sub-observation. This process yields a set of localized test-time environments $\{\mathcal{E}^{(i)}\}_{i=1}^n$, where

$$\mathcal{E}^{(i)} = (\mathcal{S}^{(i)}, \mathcal{A}, \mathcal{T}, \mathcal{R}^{(i)}). \quad (3)$$

Each localized test-time environment restricts the state and decision context while preserving the original action space and transition dynamics. Since it contains less irrelevant information and lower decision ambiguity, its local reward signal can be estimated more reliably. TTED therefore optimizes

$$\pi_{\text{TTED}}^* = \arg \max_{\pi} \sum_{i=1}^n \mathbb{E}_{\tau^{(i)} \sim \pi} [\mathcal{R}^{(i)}(\tau^{(i)})]. \quad (4)$$

By accumulating learning gains across these localized test-time environments, TTED replaces difficult direct optimization in the original complex environment with more tractable local optimization. The resulting local policy improvements are subsequently composed to solve the full task.

Test-Time Learning for LLMs

Test-time learning updates a model during inference using signals available from test instances, without ground-truth labels. For LLM agents, such signals may come from self-consistency (Zuo et al. 2025), self-assessment (Zheng et al. 2023), or external feedback (Pan et al. 2024; Chae et al. 2025). Given a pretrained policy π_{θ_0} and test instances $\mathcal{X}$, adaptation is written as

$$\pi_{\theta'} \leftarrow \mathcal{F}(\pi_{\theta_0}; \mathcal{X}), \quad (5)$$

where $\mathcal{F}$ uses unlabeled or weakly supervised inference-time signals. TTED instantiates $\mathcal{F}$ through either ICL or RL.

In-Context Learning (ICL) ICL (Brown et al. 2020; Dong et al. 2024) adapts behavior without updating θ_0 . TTED collects successful state-action pairs during trial steps as a demonstration set $\mathcal{C} = \{(s_j, a_j)\}_{j=1}^m$, and conditions subsequent action generation on this self-generated experience:

$$\pi_{\text{ICL}}(a_t | s_t) = \pi_{\theta_0}(a_t | s_t, \mathcal{C}). \quad (6)$$

This query-level adaptation reuses verified local successes when acting in the original complex environment.

Reinforcement Learning (RL) For multi-turn web tasks, TTED collects trajectories across interactions and updates the model parameters. Because ground-truth step rewards are unavailable at test time, r_t is replaced by an assessed reward r_t^* . The reward can usually be derived by matching-based self-consistency (Zuo et al. 2025) or generative reward models (Zheng et al. 2023; Liu et al. 2025c). Using undiscounted REINFORCE (Williams 1992), we estimate

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) r_t^* \right]. \quad (7)$$

The adapted parameters are then used for final execution in the full environment. The following methodology section specifies how TTED constructs r_t^* from environment decomposition and action assessment.

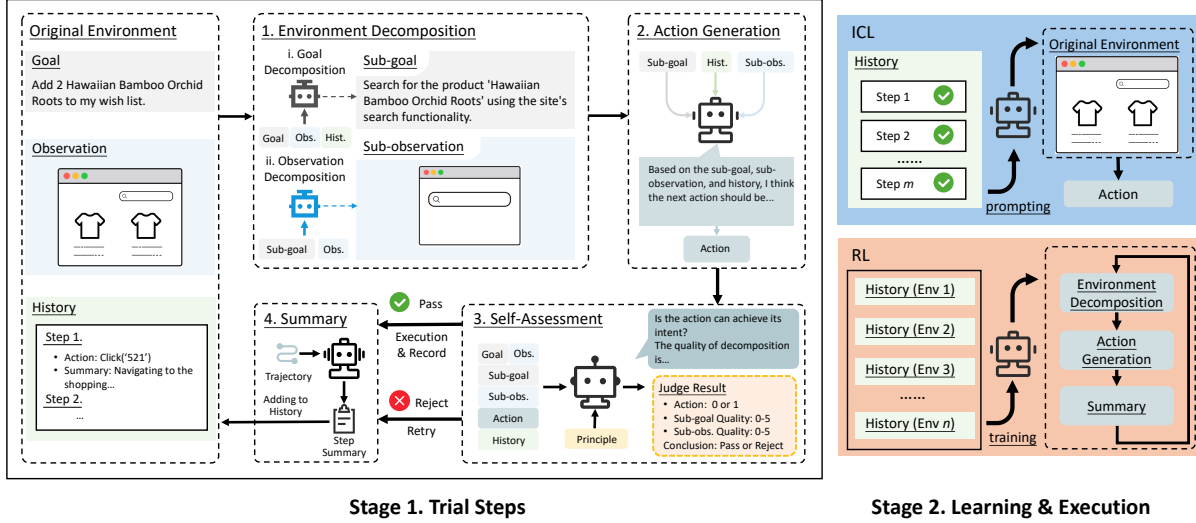


Figure 3: The Test-Time Environment Decomposition (TTED) methodology, illustrating the two-stage process of (1) Trial Steps with environment decomposition and (2) Test-Time Learning and Execution, utilizing either ICL or RL. The agent gathers experience within the decomposed sub-environments through trials. And ICL leverages correct past actions as task-specific demonstrations to solve tasks within single-turn, static webpages, while RL optimizes both decomposition process and action generation policy with trajectories from multiple environments, enabling adaptation in multi-turn, realistic web environments.

Compositional Generalization Gaps in LLMs

Compositional generalization asks whether a model can solve novel combinations of familiar components, including interface motifs, interaction primitives, layouts, and instruction patterns. To examine this gap in web agents, we extend CompWoB (Furuta et al. 2024), which composes ten base web modules but contains few tasks exceeding five components. Our extension, *CompWoB+*, uses an automated synthesis engine while retaining the original reward rules. We evaluate Qwen3-8B (Yang et al. 2025a) and GPT-4o-mini (Hurst et al. 2024) on 100 tasks per composition size from 3 to 8, yielding 600 tasks in total. Further details appear in Appendix D.

As shown in Figure 2, increasing the number of components from 3 to 8 reduces the success rate of Qwen3-8B from 85% to 19% and that of GPT-4o-mini from 97% to 46%. Because these tasks recombine familiar base modules under the original reward rules, the decline is consistent with a compositional generalization gap when components co-occur in a single observation. With TTED, the corresponding success rates at eight components increase to 61% and 88%, motivating the framework introduced next.

TTED: Test-Time Environment Decomposition

As shown in Figure 3, TTED first gathers self-assessed experience through trial steps in decomposed sub-environments, and then adapts the agent through query-level ICL or task-level RL before final execution in the original environment. All prompt templates are provided in Appendix E.

Experience Gathering via Trial Steps

To maintain a coherent reasoning trace throughout the entire trial process, TTED uses the same LLM across the four

trial steps, adopts a higher sampling temperature for diverse exploration, and represents web content as an unfolded accessibility tree following Yang et al. (2025b).

Step 1: Environment Decomposition In the first step, the agent performs decomposition to the environment state s_t (Equation (1)) to obtain a sub-environment along with an aligned goal structure. Given $s_t = (g, o_t, H_{t-1})$, TTED first predicts a localized sub-goal

$$g_t^{(i)} = \pi_{\text{goal}}(g, o_t, H_{t-1}),$$

and then extracts its task-relevant sub-observation

$$o_t^{(i)} = \pi_{\text{obs}}(o_t, g_t^{(i)}).$$

Together, they define the sub-environment state

$$S^{(i)} = (g_t^{(i)}, o_t^{(i)}, H_{t-1}),$$

while preserving the original action space and transition dynamics. For interactive websites, this sub-environment is a textual view of the original environment; for static webpages, it is instantiated in an isolated simulator.

Step 2: Action Generation Once the sub-environment is constructed, the action agent predicts the next action based on the sub-environment state. Formally, the action generation policy π_{action} produces an action tailored to the sub-goal:

$$a_t^{(i)} = \pi_{\text{action}}(g_t^{(i)}, o_t^{(i)}, H_{t-1}).$$

Step 3: Self-Assessment The agent evaluates both the quality of the environment decomposition and the correctness of the generated action as a generative reward model, following a predefined set of principles (Liu et al. 2025c).

Method	#Sub-Envs	WebArena					WorkArena
		CMS	Map	Shopping	Gitlab	Overall	
Qwen3-8B	0	12.1	8.1	14.9	12.3	12.1	18.2
SFT w/ GT	3200	11.4	12.1	17.8	14.0	14.1	17.9
RL w/ GT	3200	12.1	15.3	17.8	13.2	14.8	17.3
TTRL	3200	11.4	11.3	15.5	15.8	13.5	-
TTT	SFT	3200	10.7	9.7	12.1	12.3	11.2
	RL	3200	10.7	10.5	13.8	14.9	12.5
TTED	SFT	3200	17.4	12.1	20.7	19.3	17.6
	RL	800	19.5	15.3	21.8	20.2	-
	RL	3200	22.8	12.9	24.7	21.9	21.0

Table 1: Experimental Results for Test-Time Training on WebArena. Success rates (%) are reported for WebArena and WorkArena (out-of-distribution). “#Sub-Envs” denotes the number of sub-environments the agent has explored on during test-time learning.

The agent assigns scores to the sub-goal and sub-observation, yielding $r_{\text{goal}}, r_{\text{obs}} \in [0, 5]$, and determines the action reward with a binary score $r_{\text{action}} \in \{0, 1\}$. Formally:

$$(r_{\text{goal}}, r_{\text{obs}}, r_{\text{action}}) = \mathcal{D}_{\text{judge}}(o_t, g_t^{(i)}, o_t^{(i)}, a_t^{(i)}, H_{t-1}, g).$$

If the decomposition scores exceed a predefined threshold and $r_{\text{action}} = 1$, the trial step is marked as “Pass” and the experience (trajectory τ) is recorded. Otherwise, it is marked “Reject” and the step is retried up to a fixed limit. The self-assessment scores are used for the subsequent learning stage:

$$r_t = (r_{\text{goal}}, r_{\text{obs}}, r_{\text{action}})$$

e.g., in Equation (6) and Equation (7).

Step 4: Summary If the trial passes, the agent synthesizes a step summary to serve as a natural-language description of the executed action and its intent: $h_t = \mathcal{D}_{\text{summary}}(a_t^{(i)})$, where $\mathcal{D}_{\text{summary}}$ denotes the summary-generation function, which will not be optimized in the learning stage of TTED. This updated history $H_t = H_{t-1} \cup \{h_t\}$ manages modest context lengths, informing subsequent steps.

The environment then transitions according to $s_{t+1} = \mathcal{T}(s_t, a_t^{\text{exec}})$, where the executed action $a_t^{\text{exec}} = a_t^{(i)}$ if receiving pass signal from self-assessment, and $a_t^{\text{exec}} = \text{None}$ otherwise, indicating that no action is applied to the environment. The whole process repeats until a terminal condition is reached or the interaction step limit is exceeded.

Test-Time Learning from Experience

Specific Designs for Test-Time ICL

Self-Assessment with Executable Scripts For single-turn static webpages, pure LLM self-assessment may produce false positives. To enhance reliability, the self-assessment step leverages executable code scripts (e.g., via Selenium) under the decomposed sub-environment to execute and verify the generated action in an isolated simulation, which provides a highly accurate deterministic reward signal r_{action} for ICL without causing any side effects.

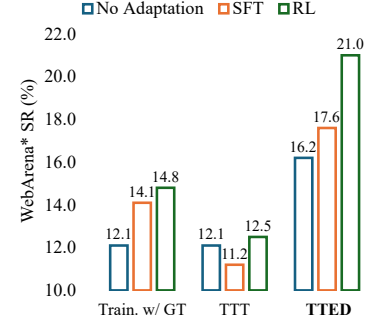


Figure 4: Task success rates (%) of test-time learning methods on WebArena (*subset). “No Adaptation” denotes the base model within the same inference pass.

ICL Demonstration Curation To ensure the quality of the test-time adaptation on the given task query, TTED strictly curates the execution history from the same task. Only trial steps that are explicitly passed by self-assessment (i.e., $r_{\text{action}} = 1$) are retained as successful demonstrations $C = \{(s_j, a_j)\}_{j=1}^m$. Failed trial steps are discarded. The policy is then conditionally updated to generate final actions based on this high-quality augmented context according to Equation (6). The full process is illustrated in Appendix D.

Specific Designs for Test-Time RL

Self-Assessment for Multi-turn Execution In realistic multi-turn websites, faithfully replicating and simulating the sub-environment is infeasible, while an incorrect action may irreversibly derail the task. Meanwhile, using LLM as a generative reward model allows TTED to flexibly assess the quality of both the decomposition and the action, where the matching-based self-consistency reward design in TTRL (Zuo et al. 2025) only applies to label actions. After test-time RL, the agent re-executes the same decomposition and action-generation pipeline without self-assessment or retry, which allows the agent to directly apply its learned decomposition and action generation strategies.

Learning from Rejected Trials Unlike ICL, RL retains both passed and rejected trials. This improves data efficiency when correct trajectories are sparse; moreover, because $r_{\text{action}} = 0$ need not imply incorrect goal or observation decomposition, the decoupled rewards allow TTED to learn useful decomposition behavior from rejected actions.

Policy Optimization Algorithm Design To optimize the unified model parameters θ governing both the decomposition process ($\pi_{\text{goal}}, \pi_{\text{obs}}$) and the action generation policy (π_{action}), we design an off-policy training scheme. Because the rollouts are generated by an older behavioral policy in the inference engine $\pi_{\theta_{\text{old}, \text{inf}}}$ during the trial steps, we utilize rollout importance sampling (Liu et al. 2025a) with the ratio $\rho_t = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}, \text{inf}}}(a_t|s_t)}$. To construct meaningful (avoiding zeros) and less variant gradients, the raw assessment scores are normalized into a bounded return $r_t^* \in [-1, 1]$. To sta-

bilize the highly variable test-time training process across diverse web layouts, we adopt the advantage normalization technique from REINFORCE++ (Hu et al. 2025). The advantage $\hat{A}_t$ is calculated by standardizing the cumulative returns across the training mini batch of collected trial environments (not necessarily from a group of trajectories from the same environment): $\hat{A}_t = \frac{r_t - \mu_{r^*}}{\sigma_{r^*} + \epsilon}$, where μ_{r^*} and σ_{r^*} are the mean and standard deviation of the normalized returns, and ϵ is a small constant for numerical stability. Inheriting from Equation (7), the final surrogate objective to be maximized is therefore formulated as:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta, \text{old}, \text{inf}}} \left[\sum_{t, i} \sum_{a^* \in \{g_t^{(i)}, o_t^{(i)}, a_t^{(i)}\}} \rho_t \nabla_{\theta} \log \pi_{\theta}(a^* | s_t) \hat{A}_t \right], \quad (8)$$

where environment decomposition and action generation are all counted in the same interaction step. We also empirically observe that this normalization is key to stabilizing advantage estimation and improving off-policy RL scalability.

Experiments and Analysis

We perform several experiments to evaluate the effectiveness of TTED and to answer a few critical research questions: **RQ1:** *Can TTED mitigate compositional generalization failures beyond synthetic web environments?* **RQ2:** *Which decomposition components are essential for effective label-free test-time learning?* **RQ3:** *What is the key factor that affects the scalability and robustness of TTED?*

Experiment Settings

Benchmarks We evaluate query-level ICL on *CompWoB+* and task-level RL on *WebArena* (Zhou et al. 2024) and *WorkArena* (Drouin et al. 2024; Boisvert et al. 2024). *CompWoB+* provides controlled compositions of web components, whereas *WebArena* and *WorkArena* contain heterogeneous, real-world interfaces. RL adapts once on *WebArena* and is evaluated both on the same benchmark distribution and on *WorkArena* without further training. We use raw DOM observations for *CompWoB+* and accessibility-tree observations through BrowserGym for the realistic benchmarks.

Baselines and Evaluation Protocol To evaluate TTED as a *label-free, transductive test-time adaptation* method, not general post-training, we experiment consistently with prior test-time learning methods (Akyurek et al. 2025; Zuo et al. 2025). On realistic web automation benchmarks, we compare TTED to a range of test-time learning baselines that do not explicitly decompose the environment:

- **No Adaptation:** The base LLM is directly applied to the full environment without any test-time adaptation.
- **Training w/ GT:** LLMs are trained with ground-truth labels from embedded rules on the full environment.
- **Test-Time Training (TTT)** (Akyurek et al. 2025): It represents a method similar to TTED but without environment decomposition.

Method	Reward Labeling	#Steps	Reward Consistency			
			r_{goal}	r_{obs}	r_{action}	Overall
TTRL	self-consistency	4.05	-	-	0.68	0.68
TTT	principled GRM	7.18	-	-	0.71	0.71
TTED	principled GRM	8.23	0.81	0.87	0.86	0.85
+ TTRL (action)	self-consistency	8.44	-	-	0.80	0.80

Table 2: Consistency evaluation (normalized AD) of reward labeling before model training from test-time learning methods with humans. Evaluation is on 82 tasks from WebArena.

Advantage Estimation	#Sub-Envs	Train. w/ GT	TTT	TTED
SFT	3200	14.1	11.2	17.6
REINFORCE	800	-	12.5	19.4
w/ Advantage	1600	-	13.5	18.4
Normalization	2400	-	13.2	20.0
	3200	14.8	12.5	21.0
REINFORCE	3200	13.0	14.3	17.6
GRPO	3200	-	13.5	17.8

Table 3: Test-time scaling experiments for test-time learning methods on WebArena with different advantage estimation strategies. We report task success rates (%) under different training budgets, i.e. numbers of explored sub-environments.

- **Test-Time RL (TTRL)** (Zuo et al. 2025): TTT method that applies self-consistency reward labeling and GRPO algorithm (Shao et al. 2024).

Orthogonal to these baselines, we also include various advantage estimation designs, including Supervised Fine-Tuning (SFT), REINFORCE (without advantage normalization), and GRPO (required by TTRL), to ablate the impact of policy optimization algorithm.

Evaluation is performed using task success rates. We also record efficiency measures such as numbers of interaction steps, token consumption, and training time. For test-time training methods, we control the training budget to ensure fair comparison. Since **constructing true local labels at step for realistic websites is non-trivial**, we also include human evaluation to examine the adaptation process of TTED, such as the quality of inferred sub-environments and the consistency between local action decisions and global task goal.

Implementation Details On *CompWoB+*, we evaluate Qwen3-8B (Yang et al. 2025a) with reasoning CoT and GPT-4o-mini (Hurst et al. 2024). On *WebArena* and *WorkArena*, we evaluate the open-source Qwen3-8B for model training. Full settings are provided in Appendix D.

Main Results

Mitigating Synthetic Compositional Generalization Failures Figure 2 illustrates performance on the synthetic *CompWoB+* benchmark as the number of concurrent environment components scales from 3 to 8. As stated in the Preliminaries section, without adaptation, both base models suffer a severe performance collapse as interface complexity increases. However, applying TTED effectively arrests this degradation. After TTED with ICL, Qwen3-8B maintains 80%+ success rates for less than 6 components, achieving

Method	WebArena
TTED	21.0
- Sub-goal decomposition	12.4
- Sub-observation decomposition	15.0
- Both = TTT	12.5

Table 4: Ablation of TTED components on WebArena, by removing sub-goal and sub-observation decomposition. We report task success rates (%) under the same RL budget.

61% at 8 components, while GPT-4o-mini retains a high score of 88%. **Takeaway:** The performance drop in cluttered environments is primarily a failure of compositional generalization rather than lacking foundational knowledge, and test-time environment decomposition provides high-quality demonstrations for ICL, which successfully preserve agent capabilities as structural complexity scales.

Performance on Realistic Web Environments As shown in Table 1 and Figure 4, TTED significantly outperforms all baselines on WebArena. Without adaptation, the base Qwen3-8B achieves only a 12.1% success rate. Standard post-training with ground-truth (GT) labels yields marginal gains (14.8% for RL) potentially because of reward sparsity, while monolithic test-time training methods (TTT, TTRL) fail to surpass these GT baselines due to noisy reward labeling in complex interfaces. Conversely, TTED (RL) achieves superior performance with 1/4 of the training budget, and further scales up to a state-of-the-art 21.0% success rate with full budget. **Takeaway:** Label-free adaptation via decomposed simple test-time environments effectively surpasses monolithic methods both in SFT and RL, even surpassing full ground-truth supervision.

Out-of-Distribution Generalization To assess cross environment transferability, we evaluated the adapted models on WorkArena without further training (Table 1). TTED (RL) maintains superior performance, achieving a 22.7% success rate compared to the base model and standard TTT. **Takeaway:** The decomposition and interaction policies learned by TTED represent robust, generalizable capabilities.

Additional model evaluations, significance tests, and efficiency analyses are provided in Appendix B.

Ablation Studies

Importance of Decomposition Components Table 4 isolates the individual contributions of the environment decomposition step in TTED. Removing sub-goal decomposition degrades the success rate severely from 21.0% to 12.4%, indicating agents disoriented without localized objectives. Removing sub-observation partitioning drops performance to 15.0%. Removing both reduces the framework to standard TTT (12.5%). **Takeaway:** The joint application of structural and goal decomposition primarily drives TTED’s efficacy.

Self-Assessment Quality Reliable reward signals are a known bottleneck in label-free test-time learning. TTED optimizes both decomposition and action policies. Without LLM-as-a-judge, e.g., TTRL, reward signals cannot cover

the trajectory. Table 2 evaluates human agreement with reward labeling prior to model training. For human evaluation, two graduate-level annotators follow guidelines (see Appendix E) and reaches 0.85 Cohen’s κ . Standard TTRL (self-consistency) and TTT (principled GRM) achieve moderate overall agreements of 0.68 and 0.71, respectively. By contrast, standard TTED reaches an exceptional 0.85. Self-consistency labeling also demonstrates improved quality when performed in decomposed environments (TTED + TTRL (action)). **Takeaway:** LLM self-assess as a GRM with proper principles yields high-fidelity signals across goal alignment, observation relevance, and action correctness, which is critical for scalable test-time learning.

RL Algorithm Design and Test-Time Scaling Table 3 investigates test-time compute scaling and policy optimization designs. As the sub-environment budget scales from 800 to 3200, standard TTT plateaus at 13.5% due to adaptation drift from noisy global updates. In contrast, TTED scales consistently up to 21.0%. Crucially, advantage normalization in sample batches stabilizes training, compared to standard REINFORCE or GRPO with peak performance to 17.8%. **Takeaway:** Cross-environment advantage normalization is essential for stabilizing and scaling off-policy test-time RL.

Qualitative Observations and Failure Cases

We also perform qualitative case analysis to understand how TTED succeeds or fails in different scenarios (see Appendix C for detail). In summary, qualitative analyses show that environment decomposition mitigates interface clutter by isolating functional components and shifting agents from brute-force interaction toward appropriate tool use. Test-time training further improves robustness to layout variations. Remaining failures usually stem from action-space misalignment or overthinking rather than incorrect decomposition; for example, agents may click to “extract” text instead of terminating with an already identified answer. Thus, TTED reduces observational complexity, while action generation remains sensitive to literal goal interpretations and environment dynamics beyond the base model’s knowledge.

Conclusion

We introduced TTED, a label-free framework that learns simple test-time environments against compositional environment complexity. Across synthetic and realistic web benchmarks, TTED improves performance for both in-domain adaptation and cross-environment transfer. Our analyses further identify self-assessment quality and algorithm designs as key factors to effective test-time scaling. Ultimately, our findings establish that equipping LLM agents with environment decomposition skills is a critical step toward inference-time scalable real-world web automation.

Ethical Statement

The deployment of robust LLM web agents has significant potential to streamline digital workflows. By addressing the compositional generalization gap—a key barrier to transferring agents from controlled benchmarks to cluttered real-world interfaces—TTED could support more reliable au-

onomous digital assistants. Its label-free test-time adaptation also increases the potential for long-horizon behaviors that scale at inference time. Nevertheless, we advocate for the strictly monitored use of such autonomous agents. TTED remains a foundational prototype rather than a production-ready system, and broader safeguards are required against digital security risks and automated misuse.

References

- Abedsoltan, A.; Zhang, H.; Wen, K.; Lin, H.; Zhang, J.; and Belkin, M. 2025. Task Generalization with Autoregressive Compositional Structure: Can Learning from $\$D\$$ Tasks Generalize to $\$D^*T\$$ Tasks? In *Forty-second International Conference on Machine Learning*.
- Akyürek, E.; Damani, M.; Zweiger, A.; Qiu, L.; Guo, H.; Pari, J.; Kim, Y.; and Andreas, J. 2025. The Surprising Effectiveness of Test-Time Training for Few-Shot Learning. In Singh, A.; Fazel, M.; Hsu, D.; Lacoste-Julien, S.; Berkenkamp, F.; Maharaj, T.; Wagstaff, K.; and Zhu, J., eds., *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 942–963. PMLR.
- Boisvert, L.; Thakkar, M.; Gasse, M.; Caccia, M.; de Chezelles, T. L. S.; Cappart, Q.; Chapados, N.; Lacoste, A.; and Drouin, A. 2024. WorkArena++: Towards Compositional Planning and Reasoning-based Common Knowledge Work Tasks. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Brown, T. B.; Mann, B.; Ryder, N.; Subbiah, M.; Kaplan, J.; Dhariwal, P.; Neelakantan, A.; Shyam, P.; Sastry, G.; Askell, A.; Agarwal, S.; Herbert-Voss, A.; Krueger, G.; Henighan, T.; Child, R.; Ramesh, A.; Ziegler, D. M.; Wu, J.; Winter, C.; Hesse, C.; Chen, M.; Sigler, E.; Litwin, M.; Gray, S.; Chess, B.; Clark, J.; Berner, C.; McCandlish, S.; Radford, A.; Sutskever, I.; and Amodei, D. 2020. Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems, NIPS '20*. Red Hook, NY, USA: Curran Associates Inc. ISBN 9781713829546.
- Chae, H.; Kim, N.; iunn Ong, K. T.; Gwak, M.; Song, G.; Kim, J.; Kim, S.; Lee, D.; and Yeo, J. 2025. Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation. In *The Thirteenth International Conference on Learning Representations*.
- de Chezelles, T. L. S.; Gasse, M.; Lacoste, A.; Caccia, M.; Drouin, A.; Boisvert, L.; Thakkar, M.; Marty, T.; Assouel, R.; Shayegan, S. O.; Jang, L. K.; Lù, X. H.; Yoran, O.; Kong, D.; Xu, F. F.; Reddy, S.; Neubig, G.; Cappart, Q.; Salakhutdinov, R.; and Chapados, N. 2025. The BrowserGym Ecosystem for Web Agent Research. *Transactions on Machine Learning Research*. Expert Certification.
- DeepSeek-AI. 2025. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(7952): 633–638.
- Deng, X.; Gu, Y.; Zheng, B.; Chen, S.; Stevens, S.; Wang, B.; Sun, H.; and Su, Y. 2023. Mind2Web: Towards a Generalist Agent for the Web. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Dong, Q.; Li, L.; Dai, D.; Zheng, C.; Ma, J.; Li, R.; Xia, H.; Xu, J.; Wu, Z.; Chang, B.; Sun, X.; Li, L.; and Sui, Z. 2024. A Survey on In-context Learning. In Al-Onaizan, Y.; Bansal, M.; and Chen, Y.-N., eds., *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 1107–1128. Miami, Florida, USA: Association for Computational Linguistics.
- Drouin, A.; Gasse, M.; Caccia, M.; Laradji, I. H.; Del Verme, M.; Marty, T.; Vazquez, D.; Chapados, N.; and Lacoste, A. 2024. WorkArena: How Capable are Web Agents at Solving Common Knowledge Work Tasks? In Salakhutdinov, R.; Kolter, Z.; Heller, K.; Weller, A.; Oliver, N.; Scarlett, J.; and Berkenkamp, F., eds., *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 11642–11662. PMLR.
- Erdogan, L. E.; Furuta, H.; Kim, S.; Lee, N.; Moon, S.; Anumanchipalli, G.; Keutzer, K.; and Gholami, A. 2025. Plan-and-Act: Improving Planning of Agents for Long-Horizon Tasks. In *Forty-second International Conference on Machine Learning*.
- Furuta, H.; Matsuo, Y.; Faust, A.; and Gur, I. 2024. Exposing Limitations of Language Model Agents in Sequential-Task Compositions on the Web. *Transactions on Machine Learning Research*.
- He, H.; Yao, W.; Ma, K.; Yu, W.; Dai, Y.; Zhang, H.; Lan, Z.; and Yu, D. 2024. WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 6864–6890. Bangkok, Thailand: Association for Computational Linguistics.
- He, Z.; Liu, Z.; Li, P.; Fung, Y. R.; Yan, M.; Zhang, J.; Huang, F.; and Liu, Y. 2025. Advancing Language Multi-Agent Learning with Credit Re-Assignment for Interactive Environment Generalization. In *Second Conference on Language Modeling*.
- Hong, S.; Zhuge, M.; Chen, J.; Zheng, X.; Cheng, Y.; Wang, J.; Zhang, C.; Wang, Z.; Yau, S. K. S.; Lin, Z.; Zhou, L.; Ran, C.; Xiao, L.; Wu, C.; and Schmidhuber, J. 2024. MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. In *The Twelfth International Conference on Learning Representations*.
- Hu, J.; Liu, J. K.; Xu, H.; and Shen, W. 2025. Reinforce++: Stabilizing critic-free policy optimization with global advantage normalization. *arXiv preprint arXiv:2501.03262*.
- Hurst, A.; Lerer, A.; Goucher, A. P.; Perelman, A.; Ramesh, A.; Clark, A.; Ostrow, A.; Welihinda, A.; Hayes, A.; Radford, A.; et al. 2024. GPT-4o System Card. *arXiv preprint arXiv:2410.21276*.
- Jaech, A.; Kalai, A.; Lerer, A.; Richardson, A.; El-Kishky, A.; Low, A.; Helyar, A.; Madry, A.; Beutel, A.; Carney, A.; et al. 2024. OpenAI o1 System Card. *arXiv preprint arXiv:2412.16720*.
- Kerboua, I.; Shayegan, S. O.; Thakkar, M.; Lù, X. H.; Boisvert, L.; Caccia, M.; Espinas, J.; Aussem, A.; Eglín,

- V.; and Lacoste, A. 2025. Focusagent: Simple yet effective ways of trimming the large context of web agents. *arXiv preprint arXiv:2510.03204*.
- Kwon, W.; Li, Z.; Zhuang, S.; Sheng, Y.; Zheng, L.; Yu, C. H.; Gonzalez, J.; Zhang, H.; and Stoica, I. 2023. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles, SOSP '23*, 611–626. New York, NY, USA: Association for Computing Machinery. ISBN 9798400702297.
- Lee, N.; Erdogan, L. E.; John, C. J.; Krishnapillai, S.; Mahoney, M. W.; Keutzer, K.; and Gholami, A. 2026. Agentic test-time scaling for webagents. *arXiv preprint arXiv:2602.12276*.
- Liu, E. Z.; Guu, K.; Pasupat, P.; Shi, T.; and Liang, P. 2018. Reinforcement Learning on Web Interfaces using Workflow-Guided Exploration. In *International Conference on Learning Representations (ICLR)*.
- Liu, J.; Li, Y.; Fu, Y.; Wang, J.; Liu, Q.; and Shen, Y. 2025a. When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch.
- Liu, K.; Lei, X.; Wang, Z.; Li, P.; and Liu, Y. 2025b. Agent-Environment Alignment via Automated Interface Generation. *arXiv preprint arXiv:2505.21055*.
- Liu, Z.; Wang, P.; Xu, R.; Ma, S.; Ruan, C.; Li, P.; Liu, Y.; and Wu, Y. 2025c. Inference-time scaling for generalist reward modeling. *arXiv preprint arXiv:2504.02495*.
- Lù, X. H.; Kazemnejad, A.; Meade, N.; Patel, A.; Shin, D.; Zambrano, A.; Stanczak, K.; Shaw, P.; Pal, C.; and Reddy, S. 2025. AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories. In *Second Conference on Language Modeling*.
- OpenAI. 2026a. Introducing GPT-5.4.
- OpenAI. 2026b. Introducing GPT-5.5.
- Pan, J.; Zhang, Y.; Tomlin, N.; Zhou, Y.; Levine, S.; and Suhr, A. 2024. Autonomous Evaluation and Refinement of Digital Agents. In *First Conference on Language Modeling*.
- Reimers, N.; and Gurevych, I. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In Inui, K.; Jiang, J.; Ng, V.; and Wan, X., eds., *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 3982–3992. Hong Kong, China: Association for Computational Linguistics.
- Selenium. 2026. Selenium Webdriver.
- Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y.; Wu, Y.; et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Shen, J.; Bai, H.; Zhang, L.; Zhou, Y.; Setlur, A.; Tong, S.; Caples, D.; Jiang, N.; Zhang, T.; Talwalkar, A.; and Kumar, A. 2025. Thinking vs. Doing: Improving Agent Reasoning by Scaling Test-Time Interaction. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.
- Sheng, G.; Zhang, C.; Ye, Z.; Wu, X.; Zhang, W.; Zhang, R.; Peng, Y.; Lin, H.; and Wu, C. 2025. HybridFlow: A Flexible and Efficient RLHF Framework. In *Proceedings of the Twentieth European Conference on Computer Systems, EuroSys '25*, 1279–1297. New York, NY, USA: Association for Computing Machinery. ISBN 9798400711961.
- Shi, T.; Karpathy, A.; Fan, L.; Hernandez, J.; and Liang, P. 2017. World of Bits: An Open-Domain Platform for Web-Based Agents. In Precup, D.; and Teh, Y. W., eds., *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 3135–3144. PMLR.
- Singh, A.; Fry, A.; Perelman, A.; Tart, A.; Ganesh, A.; El-Kishky, A.; McLaughlin, A.; Low, A.; Ostrow, A.; Ananthram, A.; et al. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267*.
- Song, Y.; Xu, F. F.; Zhou, S.; and Neubig, G. 2025. Beyond Browsing: API-Based Web Agents. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics: ACL 2025*, 11066–11085. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-256-5.
- Sun, Y.; Wang, X.; Liu, Z.; Miller, J.; Efros, A.; and Hardt, M. 2020. Test-Time Training with Self-Supervision for Generalization under Distribution Shifts. In III, H. D.; and Singh, A., eds., *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, 9229–9248. PMLR.
- Towers, M.; Kwiatkowski, A.; Balis, J. U.; Cola, G. D.; Deleu, T.; Goulão, M.; Andreas, K.; Krimmel, M.; KG, A.; Perez-Vicente, R. D. L.; Terry, J. K.; Pierré, A.; Schulhoff, S. V.; Tai, J. J.; Tan, H.; and Younis, O. G. 2025. Gymnasium: A Standard Interface for Reinforcement Learning Environments. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Wang, H.; Zou, H.; Song, H.; Feng, J.; Fang, J.; Lu, J.; Liu, L.; Luo, Q.; Liang, S.; Huang, S.; et al. 2025. Ui-tars-2 technical report: Advancing gui agent with multi-turn reinforcement learning. *arXiv preprint arXiv:2509.02544*.
- Wang, W.; Wei, F.; Dong, L.; Bao, H.; Yang, N.; and Zhou, M. 2020. MINILM: deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Proceedings of the 34th International Conference on Neural Information Processing Systems, NIPS '20*. Red Hook, NY, USA: Curran Associates Inc. ISBN 9781713829546.
- Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; brian ichter; Xia, F.; Chi, E. H.; Le, Q. V.; and Zhou, D. 2022. Chain of Thought Prompting Elicits Reasoning in Large Language Models. In Oh, A. H.; Agarwal, A.; Belgrave, D.; and Cho, K., eds., *Advances in Neural Information Processing Systems*.
- Wei, Z.; Yao, W.; Liu, Y.; Zhang, W.; Lu, Q.; Qiu, L.; Yu, C.; Xu, P.; Zhang, C.; Yin, B.; Yun, H.; and Li, L. 2025. WebAgent-R1: Training Web Agents via End-to-End Multi-Turn Reinforcement Learning. In Christodoulopoulos, C.; Chakraborty, T.; Rose, C.; and Peng, V., eds., *Proceedings of*

- the 2025 Conference on Empirical Methods in Natural Language Processing, 7909–7928. Suzhou, China: Association for Computational Linguistics. ISBN 979-8-89176-332-6.
- Williams, R. J. 1992. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine learning*, 8: 229–256.
- Xi, Z.; Chen, W.; Guo, X.; He, W.; Ding, Y.; Hong, B.; Zhang, M.; Wang, J.; Jin, S.; Zhou, E.; Zheng, R.; Fan, X.; Wang, X.; Xiong, L.; Zhou, Y.; Wang, W.; Jiang, C.; Zou, Y.; Liu, X.; Yin, Z.; Dou, S.; Weng, R.; Qin, W.; Zheng, Y.; Qiu, X.; Huang, X.; Zhang, Q.; and Gui, T. 2025. The rise and potential of large language model based agents: a survey. *Science China Information Sciences*, 68(2): 121101.
- Xu, T.; Chen, L.; Wu, D.-J.; Chen, Y.; Zhang, Z.; Yao, X.; Xie, Z.; Chen, Y.; Liu, S.; Qian, B.; Yang, A.; Jin, Z.; Deng, J.; Torr, P.; Ghanem, B.; and Li, G. 2025. CRAB: Cross-environment Agent Benchmark for Multimodal Language Model Agents. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Findings of the Association for Computational Linguistics: ACL 2025*, 21607–21647. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-256-5.
- Yang, A.; Li, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Gao, C.; Huang, C.; Lv, C.; et al. 2025a. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.
- Yang, K.; Liu, Y.; Chaudhary, S.; Fakoor, R.; Chaudhari, P.; Karypis, G.; and Rangwala, H. 2025b. AgentOccam: A Simple Yet Strong Baseline for LLM-Based Web Agents. In *The Thirteenth International Conference on Learning Representations*.
- Yao, S.; Chen, H.; Yang, J.; and Narasimhan, K. R. 2022. WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents. In Oh, A. H.; Agarwal, A.; Belgrave, D.; and Cho, K., eds., *Advances in Neural Information Processing Systems*.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K. R.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations*.
- Yuksekgonul, M.; Koceja, D.; Li, X.; Bianchi, F.; McCaleb, J.; Wang, X.; Kautz, J.; Choi, Y.; Zou, J.; Guestrin, C.; and Sun, Y. 2026. Learning to Discover at Test Time. *arXiv preprint arXiv:2601.16175*.
- Zheng, C.; Liu, S.; Li, M.; Chen, X.-H.; Yu, B.; Gao, C.; Dang, K.; Liu, Y.; Men, R.; Yang, A.; et al. 2025. Group sequence policy optimization. *arXiv preprint arXiv:2507.18071*.
- Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Zhou, S.; Xu, F. F.; Zhu, H.; Zhou, X.; Lo, R.; Sridhar, A.; Cheng, X.; Ou, T.; Bisk, Y.; Fried, D.; Alon, U.; and Neubig, G. 2024. WebArena: A Realistic Web Environment for Building Autonomous Agents. In *The Twelfth International Conference on Learning Representations*.
- Zuo, Y.; Zhang, K.; Sheng, L.; Qu, S.; Cui, G.; Zhu, X.; Li, H.; Zhang, Y.; Long, X.; Hua, E.; Qi, B.; Sun, Y.; Ma, Z.; Yuan, L.; Ding, N.; and Zhou, B. 2025. TTRL: Test-Time Reinforcement Learning. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

A Discussions, Limitations, and Additional Related Work

A.1 Discussions

Compositional Generalization based on Environment Decomposition vs. Planning We further discuss two distinct routes to achieving compositional: one through planning, which decomposes a complex goal into a sequence of sub-goals, and the other through environment decomposition, which structures the environment into simpler sub-environments. Prior work on long-horizon decision-making has predominantly followed the former paradigm, and has shown strong effectiveness across a wide range of domains (Hong et al. 2024; Yao et al. 2023; Erdogan et al. 2025). However, planning-based methods differ fundamentally from our approach of compositional generalization based on environment decomposition in several key aspects:

(1) Difference in the level of decomposition: Planning primarily operates at the level of goal decomposition, whereas TTED decomposes the entire environment, including goals, observations, and unique rewards. Formally, planning decomposes a global goal G into a sequence of sub-goals,

$$G = (g_1, g_2, \dots, g_n)$$

where each g_i represents a sub-goal to be achieved. In contrast, environment decomposition partitions the original environment into a collection of sub-environments,

$$\mathcal{E} = (\mathcal{E}^{(1)}, \mathcal{E}^{(2)}, \dots, \mathcal{E}^{(n)})$$

where each sub-environment $\mathcal{E}_i$ encapsulates its own goal, observation space, and reward function, which can be formalized as $\mathcal{E}^{(i)} = (\mathcal{S}^{(i)}, \mathcal{A}, \mathcal{T}, \mathcal{R}^{(i)})$.

(2) Difference in the optimization objective: Planning-based approaches typically decompose a complex task into a sequence of sub-goals. However, such approaches still operate in the original complex environment, where the optimization objective remains the reward of the full environment:

$$\pi_{\text{ori}}^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} [\mathcal{R}(\tau)].$$

In complex environments, estimating reward signals at test time is often expensive and inaccurate. To mitigate this issue, TTED explicitly optimizes the accumulated rewards across multiple sub-environments

$$\pi_{\text{TTED}}^* = \arg \max_{\pi} \sum_{i=1}^n \mathbb{E}_{\tau^{(i)} \sim \pi} [\mathcal{R}^{(i)}(\tau^{(i)})].$$

Through environment decomposition, we obtain simplified sub-environments with reduced complexity and clearer objectives. In such simplified environments, more accurate self-assessment reward signals can be obtained, as the agent only needs to reason over simplified sub-goals and sub-observations. This advantage is further validated through human evaluation in Ablation Study section.

In summary, these differences highlight that environment decomposition is not merely an alternative implementation of planning, but a distinct paradigm that achieves compositional generalization.

Synthetic Environment vs. Real-world Environment

Synthetic environments are typically constructed by composing multiple simpler environments into a new, more complex one (Furuta et al. 2024; Abedsoltan et al. 2025). Through such composition, these environments naturally exhibit multi-step structure and increased goal and observation complexity. A representative example is CompWoB (Furuta et al. 2024), where each environment is formed by combining several MiniWoB (Liu et al. 2018) environments. Formally, a CompWoB environment can be expressed as

$$\mathcal{E} = (\mathcal{E}_1, \mathcal{E}_2, \dots, \mathcal{E}_n)$$

where $\mathcal{E}$ denotes the composite CompWoB environment and each $\mathcal{E}_i$ corresponds to an individual MiniWoB environment. Due to their compositional construction, synthetic environments exhibit clear internal boundaries between sub-environments. As a result, environment decomposition is naturally well-defined. This property makes synthetic environments particularly suitable for studying compositional generalization under controlled settings.

In contrast, real-world environments lack such explicit compositional structure. Internal boundaries between each decomposition part are often implicit, overlapping, or ambiguous, making it difficult to obtain clear and well-defined environment decompositions. In these settings, environment decomposition cannot be trivially derived from the environment construction, but instead relies heavily on the agent capabilities of planning and reasoning to infer meaningful sub-environments. Also, for practical deployment, unsafe operations or actions with critical side effects in sub-environment exploration should be suspended and wait for user confirmation.

Consequently, although synthetic environments provide a convenient and controlled testbed for studying compositional generalization, the substantial gap between synthetic and real-world settings necessitates validating such methods in realistic environments. In real-world scenarios, compositional structure must be actively discovered rather than given by design, making it essential to evaluate whether improvements observed in synthetic benchmarks can translate to practical and complex environments.

TTED vs. Observation Pruning or Subgoal-driven RL

Observation pruning methods, such as AgentOccam (Yang et al. 2025b) and FocusAgent (Kerboua et al. 2025), aim to manage long-context inputs by truncating, filtering, or selecting relevant observations from temporal trajectories. These methods are orthogonal to TTED: they primarily reduce the context length at inference time, whereas TTED uses localized environment decomposition as *learning signals* for label-free test-time adaptation. In this sense, prior observation-reduction methods motivate the need for more effective self-assessment and adaptation mechanisms, rather than weakening our setting.

Subgoal-driven RL also differs from TTED in both motivation and usage. Methods that use checklists, principles, or subgoals for reward modeling typically aim to enhance the overall reward signal by aggregating multiple intermediate scores along a trajectory. In contrast, TTED decomposes a

complex task into a *specific sub-goal* aligned with the corresponding decomposed environment, enabling the agent to perform localized self-learning during test-time adaptation. Therefore, TTED is not designed to provide a full-trajectory outcome reward, but to construct localized learning signals that better match the current sub-environment.

A.2 Limitations

Despite the effectiveness of our approach, it has several limitations that merit discussion.

The quality of environment decomposition is inherently dependent on the capability of pre-trained models. As discussed in Appendix A.1, TTED relies on the ability of the backbone model to identify meaningful sub-environments from complex environments. When the decomposition quality is poor due to extreme incompetence of models, errors or ambiguities in the resulting sub-environments may propagate to downstream decision-making, limiting the effectiveness of compositional generalization. This issue can be pronounced in realistic settings, where environment boundaries are more implicit and more difficult to infer. Improving the robustness and reliability of environment decomposition remains an important direction for future work.

Agent self-assessment in multi-turn web environments is designed with simplicity and does not leverage sources outside the backbone model. TTED employs agent self-assessment to provide reward signals that assess action execution and decomposition quality in the test time. While this design enables flexible and scalable training without ground-truth supervision, it also introduces potential bias and noise inherited from the backbone model itself. Inaccurate or inconsistent judgments may affect training stability and limit performance gains. Utilizing larger backbone models or exploring more advanced self-rewarding approaches beyond GRM, e.g., script-based simulator for static web-pages in TTED with ICL, could help mitigate this limitation in future.

A.3 Additional Background on LLM Web Agents and Evaluation

Recent advancements have positioned LLMs as capable agents for sequential decision-making (Yao et al. 2023), and thus general web automation is implemented as a natural application domain for agents to interact with complex observation spaces, including HTML, DOM trees, accessibility metadata, screenshots, or mixed multimodal representations, and multi-level action interfaces, including low-level clicks and keystrokes, high-level API calls, or script-based code generation (He et al. 2024; Song et al. 2025; Wei et al. 2025; Yang et al. 2025b), and perform diverse tasks such as basic component manipulation, web shopping, information retrieval, and more realistic scenarios (Shi et al. 2017; Yao et al. 2022; Deng et al. 2023; Zhou et al. 2024; Drouin et al. 2024). Recent efforts have explored training LLMs with long reasoning capabilities (Jaech et al. 2024; DeepSeek-AI 2025) on interactive web environments with RL (Wei et al. 2025; Wang et al. 2025; Shen et al. 2025; OpenAI 2026a).

For evaluation, de Chezelles et al. (2025) developed a unified protocol for benchmarking web agents, while Lù et al. (2025) consider how well LLMs can judge web agent performance as reward models.

B Additional Experimental Results

B.1 CompWob+ Results with Larger LLMs

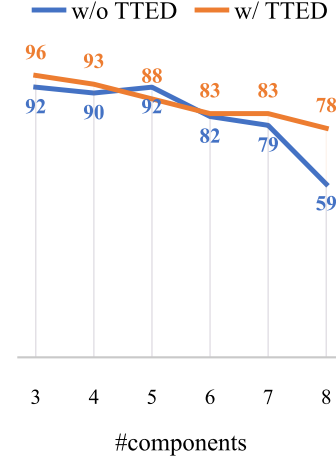


Figure 5: Task success rates (%) on CompWob+ with Qwen3 14B

To further evaluate the robustness of TTED across different models, we additionally conduct experiments on CompWoB+ using Qwen3-14B. The results are shown in Figure 5.

As model size increases, the base model itself demonstrates improved performance across all compositional tasks. However, the model still experiences a significant performance drop when the composition number reaches 8, with a 25.3% decrease compared to tasks with 7 compositions. In contrast, with TTED combined with ICL, the performance remains stable as task complexity increases, achieving 78% accuracy even for tasks with 8 compositions. These results demonstrate that TTED remains effective when applied to different models.

B.2 Webarena Results with Additional LLMs

To test 10B-scale open-sourced models with long CoT reasoning ability, we train Qwen3-14B on WebArena with RL and evaluate it using the same protocol as Qwen3-8B. The results are shown in Table 5.

Compared to bare Qwen3-14B, TTED achieves relative gains of 16.6% on undecomposed environments and 4.2% on decomposed environments, while still outperforming ground-truth RL. The smaller margin may reflect higher absolute performance and increasingly dominant bottlenecks beyond environment complexity; other baselines follow the Qwen3-8B trend.

In addition, we evaluate the performance of proprietary models on WebArena, where GPT-4o and GPT-5.5 achieve success rates of 23.5% and 67.3%, respectively. Although

Model	Method	WebArena SR
Gemma3-12B	TTI	22.1
GPT-4o	-	23.5
GPT-5.5*	-	67.3
Qwen3-8B	-	12.1
	RL w/ GT	14.8
	TTT (RL)	12.5
	TTED (RL)	21.0
Qwen3-14B	-	19.3
	RL w/ GT	20.0
	TTT (RL)	18.7
	TTED (RL)	22.5

Table 5: Success rate (SR, %) on the selected WebArena subset. Results in gray are provided as contextual references only. TTI is not a test-time training method and uses ground-truth labels in the benchmark. GPT-5.5* is evaluated on full WebArena-Verified (OpenAI 2026b), which shares similar task difficulty but uses different verifiers.

TTED does not match the performance of these proprietary models, model scale should also be taken into account. For the same Qwen3-8B backbone, TTED improves WebArena from 12.1% to 21.0% and WorkArena from 18.2% to 22.7% (73.6% and 24.7% relative improvement, respectively).

B.3 Statistic Significance Tests

We repeated the main WebArena results (Table 1, Figure 4) three times. Rollout sampling uses temperature 1.0, whereas evaluation uses temperature 0. Across runs, the absolute variation is below 0.5% and thus results are not changed. Over the 561 WebArena tasks, Qwen3-8B TTED achieves an SR of 21.0% with a 95% Wilson CI of [17.9, 24.6], compared with [9.7, 15.1] for the base model, [12.1, 18.0] for RL w/ GT, and [10.0, 15.5] for TTT. One-sided aggregate proportion tests show that TTED significantly outperforms these baselines ($p < 10^{-4}$, $p = 0.0032$, and $p < 10^{-4}$, respectively). For Qwen3-14B, the gains remain positive ($p = 0.093$, $p = 0.153$, and $p = 0.061$, respectively) (Table 5).

Table 6: Efficiency Measures of TTED-trained models on WebArena. “#Steps” denotes the average number of interaction steps in a task, and “#Prompt Tokens” denotes the average input tokens of the action generation policy.

Method	#Sub-Envs	WebArena SR (%)	#Steps	#Prompt Tokens
Qwen3-8B	0	12.1	4.72	6338.9
TTED (RL)	3200	21.0	8.29	3948.2

B.4 Efficiency Analysis

Table 6 demonstrates TTED’s context efficiency. While exploratory trial steps naturally increase the average interaction steps per task (8.29 vs. 4.72), restricting the action generation policy to partitioned sub-observations significantly reduces average prompt token consumption. In summary, TTED extends interaction scales during test-time learning, and localized observations successfully reduce per-action computational overhead, which is very critical for real-world web

automation where textual representation of websites can be six-figure token counts.

C Qualitative Studies

In this section, we provide detailed qualitative case studies to complement the observations discussed in Experiments section. Including cases from CompWeb+ (Furuta et al. 2024) and WebArena (Zhou et al. 2024).

C.1 Webarena Case Studies

To further understand the behavior of TTED, we present three representative case studies from WebArena. These cases illustrate (1) success enabled by TTED, (2) performance improvement after test-time training, and (3) failure cases where correct decomposition still leads to errors due to action-space misalignment.

Case 1: Success with TTED but failure without TTED

In this case, the agent without TTED fails due to misinterpretation of key elements in the complex environment. In contrast, the agent with TTED isolates relevant components, correctly identifies the key elements, and successfully completes the task. This case demonstrates that environment decomposition simplifies the environment, enabling agents to perform more accurate reasoning and reducing hallucinations.

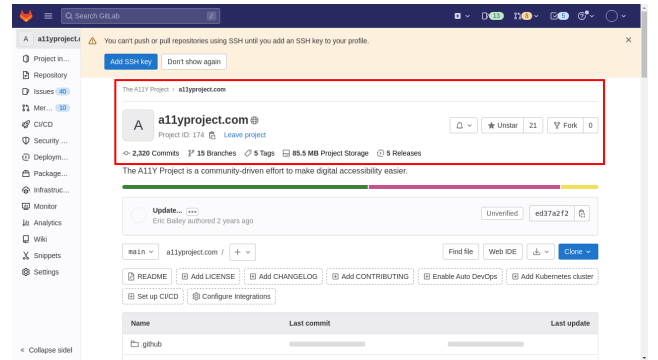


Figure 6: Observation for # Case 1.

Case 1: Success with TTED vs. Failure without TTED

Goal: How many commits did Eric make to a11yproject on 3/2?

Website: GitLab

Common Steps:

- step 1. click(“393”) # navigate to “The A11Y Project / a11yproject.com” repository page

Current Page: Repository page, see Figure 6 for detail.

Observation:

RootWebArea ‘The A11Y Project / a11yproject.com · GitLab’, focused

[67] banner ‘

[68] link ‘Skip to content’

StaticText ‘GitLab’

Customers

70 records found

Name	Email	Group	Phone	ZIP	Country	State/Province	Customer Since
Jane Doe	jane.doe@gmail.com	General	2125551212	10001	United States	New York	Apr 23, 2023 12:13:19 AM
Jane Smith	jane.smith@gmail.com	General	2065554321	98122	United States	Washington	Apr 19, 2023 5:45:24 PM
Jane Smith	janesmith456@yahoo.com	General	3105555555	90210	United States	California	Apr 19, 2023 5:45:41 PM
Jason Miller	jason.miller@yahoo.com	General	3035551212	80202	United States	Colorado	Apr 19, 2023 5:46:34 PM
Jennifer White	jennifer.white@yahoo.com	General	2137418080	90015	United States	California	Apr 19, 2023 5:46:17 PM
Jessica Chang	jessica.chang@hotmail.com	General	2065551212	98101	United States	Washington	Apr 23, 2023 12:13:26 AM
Jessica Nguyen	jessica.nguyen@gmail.com	General	2065551212	98101	United States	Washington	Apr 23, 2023 12:13:58 AM
Jessica Wong	jessica.wong@gmail.com	General	3125551212	60601	United States	Illinois	Apr 23, 2023 12:13:48 AM
John Doe	john.doe123@gmail.com	General	2125551212	10001	United States	New York	Apr 19, 2023 5:45:37 PM
John Lee	john.lee@yahoo.com	General	3125556789	60611	United States	Illinois	Apr 19, 2023 5:45:21 PM
John Smith	john.smith.ay@gmail.com	General	2068812362	35213	United States	Alabama	Apr 19, 2023 5:44:57 PM
John Smith	john.smith@yahoo.com	General	2145551212	75201	United States	Texas	Apr 23, 2023 12:13:22 AM

(a)

Customers

70 records found

Search by keyword

Name	Email	Group	Phone	ZIP	Country	State/Province	Customer Since
Adam Garcia	gaminggro56@gmail.com	General	2065555555	98122	United States	Washington	Apr 19, 2023 5:46:14 PM
Adam Garcia	adam.garcia@gmail.com	General	6025551212	85004	United States	Arizona	Apr 19, 2023 5:46:40 PM
Alex Johnson	fitnessjunkie28@yahoo.com	General	3105555555	90212	United States	California	Apr 19, 2023 5:46:07 PM
Alex Martin	alex.martin@gmail.com	General	2125551212	10001	United States	New York	Apr 19, 2023 5:46:20 PM
Alexander Thomas	alexander.thomas@hotmail.com	General	3235551212	90028	United States	California	Apr 19, 2023 5:46:50 PM
Amanda Kim	amanda.kim@gmail.com	General	2015551234	07030	United States	New Jersey	Apr 23, 2023 12:13:15 AM
Anna Nguyen	anna.nguyen@yahoo.com	General	7135551212	77002	United States	Texas	Apr 23, 2023 12:13:09 AM
Aria Brown	aria.brown94@yahoo.com	General	9160000000	90045	United States	California	Apr 19, 2023 5:46:01 PM

(b)

Figure 7: Observation for # Case 2.

[74] link 'Dashboard'
[75] image "

Divergence Point:

failure in w/o decomposition model
<think>

1. The agent think 'commit history' should be found in current page.

First, I need to look at the current web page's accessibility tree provided. . . The observation includes various elements, but the main area of interest is the commit history.

2. Then, the agent interpreted the current page as merely a file list, without identifying any elements related to commit history.

Looking at the elements, there's a table labeled 'Files, directories, and submodules in the path / for commit reference main'. . . But in the current observation, there's no explicit mention of a commit history table or a list of commits. The table here seems to list files and directories, not commits.

3. The agent identified the key element required to complete the task, but mistakenly interpreted it as plain text rather than a link, and therefore excluded it.

Wait, there's a StaticText "2,320 Commits" in the list-item under the navigation. That might indicate that there are 2,320 commits in total, but that's not specific to Eric or March 2nd.

Final Action: report_infeasible("The current page does not provide commit history filtered by author or date. Eric's commits on March 2nd cannot be determined from the available data.")

success in w/ decomposition model

- Goal Decomposition: Click the link labeled '2,320 Commits' to view the project's commit history and identify commits made by Eric on March 2nd.

- Corresponding Reasoning: <think>

1. Agent correctly interpreted the current page. First, the agent is on the project's main page.

2. Then, the agent identified the key sub-goal for the next step and proposed a potential solution.

The next step would be to check if there's a way to view commits by a specific author and date.

- On GitLab, typically, you can navigate to the repository's "Repository" section, then go to "Commits" or "Activity" to see commit history.

- Alternatively, there might be a "Commits" link in the main content. The AXTree has a list under the main area with "2,320 Commits" as a link.

3. Finally, the agent selected the second approach and ultimately formulated a concrete sub-goal. The AXTree has a link labeled "2,320 Commits" under the main area's navigation. Clicking that link would take the agent to the commit history page.

</think>

- Observation Decomposition: See the region highlighted by the red rectangle in Figure 6 for detail.

Final Action: click('683') # Correctly click the "2,320 Commits" link. ✓

Case 2: Success after training. For the same task, the agent before training adopts a brute-force strategy to scan all entries, which leads to hallucinations under complex contexts and ultimately results in failure. After training, the agent learns to take advantage of the search functionality to efficiently locate the target, leading to successful completion of the task. The key difference lies in the selection of the agent strategy and robustness under complex observations.

Case 2: Success after Training vs. Failure before Training

Goal: Find the customer name and email with phone number 2137418080

Website: CMS

Common Steps:

- step 1 & 2. Navigate to 'All customers' page

Current Page: Customers information page, see Figure 7(a) for detail.

Observation:

Accessibility Tree:

RootWebArea 'Customers / Customers / Magento Admin', focused

[175] link 'Magento Admin Panel'

[176] image 'Magento Admin Panel'

[177] navigation "

[178] menubar "", orientation='horizontal'

[180] link 'DASHBOARD'

StaticText "'

StaticText 'DASHBOARD'

[183] link 'SALES'

StaticText "'

StaticText 'SALES'

...

Divergence Point:

failure in w/o train model

- **Goal Decomposition:** Click the row containing the phone number '2137418080' to select the corresponding customer record.

- **Corresponding Reasoning:** <think>

1. The agent correctly interpreted the current page.

Okay, let's see. The task is to find the customer name and email with the phone number 2137418080. The current page is the Magento Admin's Customers section, and there's a table listing various customers with their details.

...

2. Then, the agent chose to brute-force all entries instead of using the search function, which set the stage for subsequent errors.

Looking at the AXTree, there's a table with columns like Name, Email, Phone, etc. The phone numbers are in the 'Phone' column. I need to scan through the rows to find the entry where the Phone column matches 2137418080.

3. Due to the overly complex environment, the agent exhibited hallucination while searching for the target, ultimately leading to an incorrect answer.

Wait, looking further down, there's a row [2310] where the Phone is '2137418080'. That's the one we need.

* Element [2310] corresponds to the entry with the phone number '2065551212', rather than the goal target. </think>

- **Observation Decomposition:** See the region highlighted by the red rectangle in Figure 7(a) for detail. # The agent again hallucinated, modifying the entry with

phone number '3035551212' into the supposed target entry.

* **Hallucination:** [2279] gridcell '3035551212' → [2279] gridcell '2137418080' ✗

- **Corresponding Reasoning:** <think>

... Yes, that's the row with the phone number '2137418080'. So the entire row [2268] and its children are relevant. ...

</think>

Final Action: click('2279') # The element [2279] actually is '3035551212'. ✗

success in w/ train model

- **Goal Decomposition:** Search for the phone number '2137418080' in the 'Search by keyword' textbox and click the 'Search' button to filter the customer list.

- **Corresponding Reasoning:** <think>

1. The agent correctly interpreted the current page and proposed using search to locate the target.

The current page is the Magento Admin's Customers section, and there's a table listing all customers. The user has already navigated to the Customers section, so now they need to search through the customer data.

2. Then, the agent confirms that the search box exists on the current page.

The next step should be to search for the phone number 2137418080 in the table. ... Wait, in the AXTree, there's a textbox labeled 'Search by keyword' and a button 'Search'. So maybe the agent should use that search functionality to filter the customers by phone number.

</think>

- **Observation Decomposition:** See the region highlighted by the red rectangle in Figure 7(b) for detail.

The agent successfully locates the search box.

Final Action: fill("854", "2137418080") # [854] textbox 'Search by keyword' ✓

Case 3: Failure despite correct environment decomposition. In this case, the environment decomposition is correct, but the agent still fails due to action-space misalignment. Specifically, the agent overthinks when attempting to map the sub-goal to an unavailable "extract" action, ultimately selecting an incorrect action despite already identifying the correct answer.

Case 3: Failure Caused by Overthinking

Goal: List the top 1 search terms in my store

Website: CMS

Current Page: Dashboard, see Figure 8 for detail.

Observation:

RootWebArea 'Dashboard / Magento Admin', focused

[148] link 'Magento Admin Panel'

[149] image 'Magento Admin Panel'

[150] navigation "

[151] menubar "", orientation='horizontal'

[153] link '\ue604 DASHBOARD'

Search Term	Results	Uses
tanks	23	1
nika	0	3
joust bag	10	4
hollister	1	19
Antonia Racer Tank	23	2

Search Term	Results	Uses
hollister	1	19
joust bag	10	4
Antonia Racer Tank	23	2
tanks	23	1
WP10	1	1

Figure 8: Observation for # Case 3.

StaticText ‘\ue604’
StaticText ‘DASHBOARD’
[156] link ‘\ue60b SALES’
...

Failure Point:

- **Goal Decomposition:** Extract the search term with the highest ‘Uses’ value from the ‘Top Search Terms’ table and list it as the top search term.

- **Corresponding Reasoning:** <think>

In the goal decomposition stage, the task answer has already been fully identified.

The first thing I notice is that the table under “Top Search Terms” already has entries. For example, “hollister” has 19 uses, “Joust Bag” has 4, “Antonia Racer Tank” has 2, “tanks” has 1, and “WP10” has 1. The user wants the top 1 search term, which would be the one with the highest number in the “Uses” column.

...

Since the data is already present on the page, the next logical step is to extract this information.

</think>

- **Observation Decomposition:** See the region highlighted by the red rectangle in Figure 8 for detail. # Extract all necessary elements according sub-goal correctly from original observation.

Final Action: click(“918”) # [918] gridcell ‘hollister’



- **Corresponding Reasoning:** <think>

1. The agent correctly analysis that ‘Hollister’ is the finally answer.

Looking at the provided data, the rows under ‘Top Search Terms’ include entries like ‘hollister’ with 19 uses, ‘Joust Bag’ with 4, ‘Antonia Racer Tank’ with 2, ‘tanks’ with 1, and ‘WP10’ with 1. The highest ‘Uses’ here is 19 for ‘hollister’.

2. However, the agent is misdirected by the action “extract” in the sub-goal, which does not exist in the action space. As a result, the agent starts overthinking and attempts to find the closest action to “extract.”

But the assistant’s role here is to perform actions on the webpage. Since the user is asking for the top search term, and the data is present in the table, the assistant needs to extract that information. However, the available actions don’t include reading text from elements. The actions available are things like click, fill, etc.

</think>

C.2 Compwob+ Case Studies

For static webpages such as CompWoB+, TTED can leverage successful trial steps as demonstrations for in-context learning (ICL), enabling it to directly generate final actions for a single task query. We provide a detailed description in Appendix D.3.

We present a representative case from CompWoB+ to illustrate TTED in single-turn, static webpages. In this case, the task requires completing multiple sequential operations, including reading a table, clicking a download button, and selecting a radio option. TTED first decomposes the original goal into multiple sub-goals and extracts the corresponding sub-observations for each sub-goal.

Based on the decomposed sub-goal, the agent generates an initial sequence of actions. A JavaScript-based simulation is then applied to validate each action. The simulation reveals that clicking a table cell is unnecessary and ineffective, leading to rejection by the self-assessment module. The agent subsequently retries action generation, removes redundant interactions, and refines the action sequence. After validation through self-assessment process, the revised action sequence is accepted, leading to successful task completion.

This case demonstrates how environment decomposition simplifies compositional tasks, while the action generation – self-assessment loop further improves action reliability and robustness.

Compwob+ Case: Environment Decomposition

Goal: Read the table and enter the value of Country into the text field and press Submit, and then click on the button “Download”, and then select radio button gYirC and click Submit

Page: See Figure 9. for detail.

- **Goal Decomposition:**

1. read-table: Read the table and enter the value of Country into the text field and press Submit
2. click-button: Click on the button “Download”
3. click-option: Select radio button gYirC and click Submit

- **Observation Decomposition:** See the region highlighted by rectangles in Figure 9 for detail.

- **Part of Sub-Observation:**

subtask 2: click-button: Click on the button “Download”.
html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Click Button</title>
```

```
</head>
```

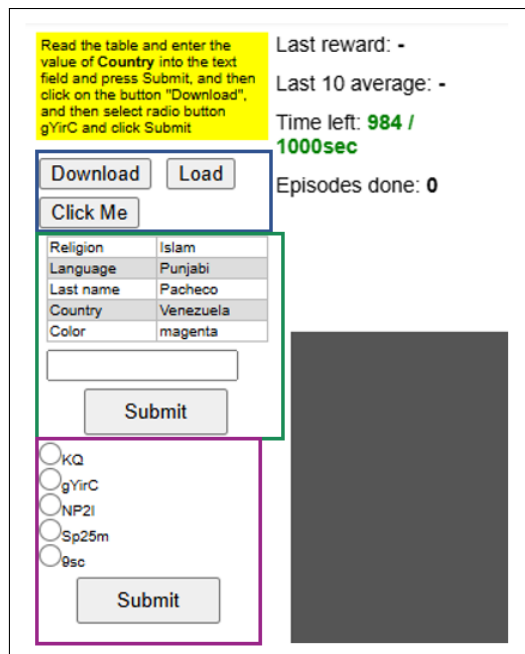


Figure 9: Observation for # Compwob+.

```
<body>
  <div id="buttonsContainer" data-wob_ref="4"
data-wob_eps="e4">
    <button class="generated-button" data-
wob_ref="5" data-wob_eps="e4">Download</button>
  </div>
</body>
</html>
```

Compwob+ Case: Trial Action Generation & Simulation & Self-Assessment

- **Sub-Goal:** Read-table: Read the table and enter the value of Country into the text field and press Submit.
- **Sub-Observation:** See the region highlighted by green rectangle in Figure 9 for detail.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Read Table</title>
```

```
</head>
```

```
<body>
```

```
  <div id="tab" data-wob_ref="8" data-
wob_eps="e4">
```

```
    <table data-wob_ref="9" data-wob_eps="e4">
```

```
      <tr data-wob_ref="10" data-wob_eps="e4">
```

```
        <td data-wob_ref="11" data-
wob_eps="e4">Religion</td>
```

```
      ...
```

```
- Action:
```

1. clickxpath //tr[td[text()='Country']]/td[2]
2. clickxpath //input[@id='tt']

3. type Venezuela

4. clickxpath //button[@id='tablesubbtn']

- **JS Simulation:**

```
// Action 1. clickxpath //tr[td[text()='Country']]/td[2]
var countryCell = document.evaluate("//tr[td[text()='Country']]/td[2]", document, null, XPathResult.FIRST_ORDERED_NODE_TYPE, null).singleNodeValue;
```

```
return countryCell === document.activeElement;
```

```
// Action 2. clickxpath //input[@id='tt']
```

```
var inputField = document.getElementById("tt");
return inputField === document.activeElement; // Verifies if the input field is focused after the click
```

```
// Action 3. type Venezuela
```

```
var inputField = document.getElementById("tt");
```

```
return inputField.value === "Venezuela";
```

```
// Action 4. clickxpath //button[@id='tablesubbtn']
```

```
var submitButton = document.getElementById("tablesubbtn");
return submitButton === document.activeElement;
```

Simulation Result: [False, True, True, True]

- **Self-Assessment Result:** Reject

- **Corresponding Reasoning:** <think>

1. Instruction 1: The XPath '//tr[td[text()='Country']]/td[2]' correctly targets the "Venezuela" cell in the table. However, the inspect result is ****False****, indicating the click action failed. This is likely because clicking a table cell (which is not an input or button) does not interact with the content (e.g., select text or copy it). Since the input field '#tt' is already pre-filled with "Venezuela", this step is ****unnecessary**** and invalidates the task flow.

2. Instruction 2: Clicking '//input[@id='tt']' is valid and focuses the input field (inspect result: ****True****).

3. Instruction 3: Typing "Venezuela" into the input field is redundant since the field is already pre-filled. However, the observation confirms the value matches, so the action is technically valid (inspect result: ****True****).

4. Instruction 4: Clicking the "Submit" button is valid (inspect result: ****True****), though it has no functional effect as per the note.

</think>

Retry Action Generation → Self-Assessment → Assessment result: **Pass**

Final Action:

1. clickxpath //input[@id='tt']
2. type Venezuela
3. clickxpath //button[@id='tablesubbtn']



Final Action Generation

Goal: Read the table and enter the value of Country into the text field and press Submit, and then click on the button "Download", and then select radio button gYirC and click Submit.

Page: See Figure 9 for detail.

Observation:

Goal: Read the table and enter the value of Country into the text field and press Submit, and then....

Observation:



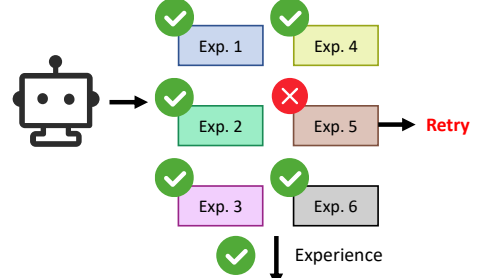
1. Environment Decomposition

- Sub-goal 1: click-button
Click on the button "Confirm".
- Sub-goal 2: login-user
Type the username "cheree" and password "beR9" and press Login.
- Sub-goal 3: edit-text
Type the following text into the editor: "In. Morbi. Aliquet est." and press submit.
- Sub-goal 4: read-table
Read the table and enter the value of Country into the text field and press Submit.
- Sub-goal 5: click-widget
Click on a "radio" widget.
- Sub-goal 6: click-option
Select radio button poYARk4 and click Submit.

2. Action

- Sub-goal 1:
Click on the button "Confirm".
Obs. : ...
Action:
clickxpath //button[text]="Confirm"]

3. Self-Assessment



4. Overcome in Original Env.

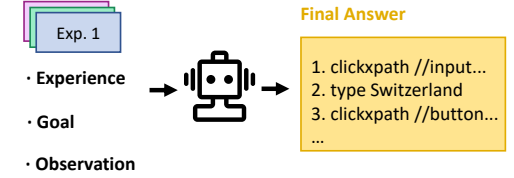


Figure 10: Overall workflow of TTED with in-context learning (ICL) on CompWoB+. Given a complex goal and the original observation, the agent first performs environment decomposition to divide the task into multiple sub-goals and extract the corresponding sub-observations. For each decomposed sub-environment, the agent generates trial actions and evaluates them through the self-assessment module. The successful trials are stored as experiences, while failed trials trigger retry until success or reaching the maximum retry limit. Finally, the collected successful experiences are leveraged as demonstrations for ICL, guiding the agent to directly generate the final action sequence in the original environment.

```
<div id="wrap" data-wob_ref="2" data-wob_eps="e4">
  <div id="query">
    Read the table and enter the value of ...
  </div>
  <div id="area" data-wob_ref="3" data-wob_eps="e4">
    ...
```

Experience:

...
Sub-goal 2: Read-table: Read the table and enter the value of Country into the text field and press Submit.
Sub-observation 2: See the region highlighted by green rectangle in Figure 9 for detail.
Action:
1. clickxpath //input[@id='tt']
2. type Venezuela
3. clickxpath //button[@id='tablesubbtn']
Sub-goal 3: ...

```
...
Final Action:
1. clickxpath //input[@id='tt']
2. type Venezuela
3. clickxpath //button[@id='tablesubbtn']
4. clickxpath //button[text()='Download']
5. clickxpath //input[@id='ch1']
6. clickxpath //button[@id='optionsubbtn']
```

D Implementation Details

The models and datasets utilized in this study are intended strictly for research purposes. Furthermore, we have maintained full adherence to all applicable licensing agreements and established ethical frameworks.

D.1 Implementation Details of Data Collection

In this section, we present dataset details, including the construction of CompWoB+ (Furuta et al. 2024) and the subset

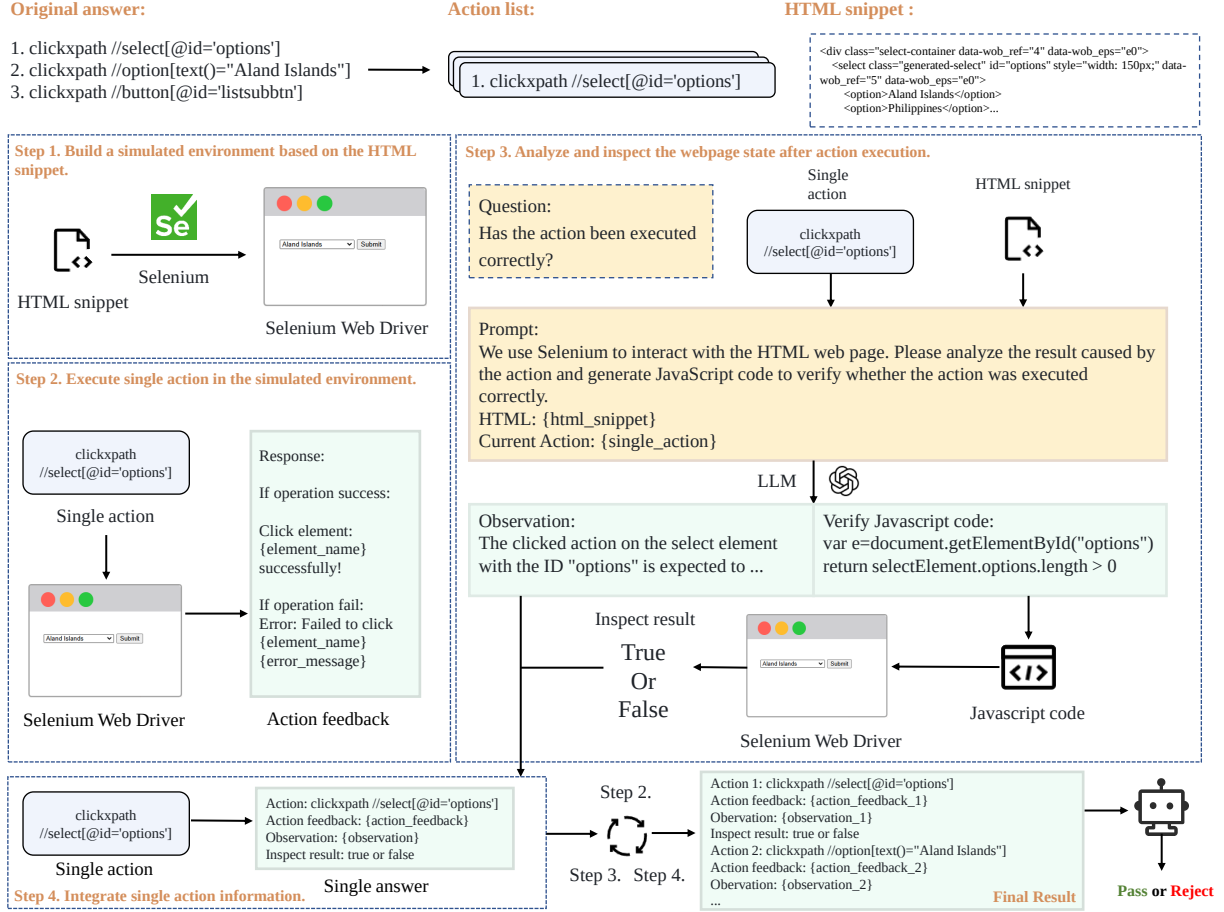


Figure 11: Workflow of the self-assessment module in TTED with ICL. For each generated action, a simulated environment is first constructed from the HTML snippet using Selenium. The action is then executed within the simulated environment, and the resulting webpage state is analyzed. A JavaScript verification script is generated to inspect whether the action is performed correctly. The execution feedback, observations, and verification results are integrated to determine whether the action is successful and returns the final judgment.

of tasks selected from WebArena (Zhou et al. 2024).

For CompWoB+, we design a task generator to address the limited number of high-composition tasks in the original dataset. The generator first selects multiple base miniwob (Liu et al. 2018) tasks according to a configuration file, which specifies both the task types and the composition size. These selected tasks are then composed into a unified web page to construct a composite environment. Each base task is implemented as an independent UI module, and for each module, the generator randomly samples a task instance. For example, a button with randomly generated text, a list with randomly sampled options, or a table with randomly generated entries may be created. All randomness is controlled by a global random seed. By varying the seed, the generator produces different task instances, enabling large-scale task generation. In summary, the generator supports arbitrary combinations of base tasks and can generate composite tasks with varying composition sizes.

Using this generator, we construct the CompWoB+

dataset. Specifically, we select 10 base tasks and use the generator to arbitrarily combine them into composite tasks. The list of base tasks is shown in Table 7. We generate composite tasks with composition sizes ranging from 3 to 8. For each composition size, we construct 10 distinct composite tasks, and for each task, we further generate 10 randomized instances for evaluation. The resulting dataset statistics are summarized in Table 8. As a result, Compwob+ ensures consistency with the difficulty of the original dataset by selecting and combining unit modules from the original dataset, and task quantity is substantially enlarged (from 50 to 600).

For WebArena, some tasks are difficult to collect and evaluate due to network-related issues. Therefore, we conduct experiments on a selected subset of tasks. The selected subset is shown in Table 9. Specifically, we use 561 of 812 WebArena tasks, and Reddit could not operate reliably under our network, whose completion rate is lower than 5%. Even after this exclusion, our benchmark remains larger than the commonly used WebArena-Lite (165 tasks).

ID	Base Task	ID	Base Task
1	click_button	6	login_user
2	click_link	7	choose_list
3	click_widget	8	read_table
4	click_checkboxes	9	edit_text
5	click_option	10	enter_time

Table 7: Base tasks used for constructing composite tasks in CompWoB+.

Composition Size	Task	Instances / Task	Total Instances
3	10	10	100
4	10	10	100
5	10	10	100
6	10	10	100
7	10	10	100
8	10	10	100
Total	60	—	600

Table 8: Statistics of composite tasks in CompWoB+.

For text-based answers in WebArena, the evaluation relies on LLM-based fuzzy matching. In the original WebArena benchmark, GPT-4o-Turbo is used for answer evaluation. To improve the matching quality, we replace it with GPT-5-mini (Singh et al. 2025). It is worth noting that this process is used only internally by the benchmark to determine whether a task is successfully completed, and is not used as a reward model to guide TTED.

	CMS	Map	Shopping	Gitlab	Reddit*	Overall
#Tasks	149	124	174	114	0	561

Table 9: Statistics of the selected WebArena task subset. *For Reddit, we remove all tasks due to network issues.

D.2 Implementation Details of Model Training

In this section, we provide the hyper-parameter settings and implementation details for test-time training of the TTED framework, as well as the baseline methods discussed in Experiments section.

Infrastructure and Base Model The training experiments are conducted using the open-source Qwen3-8B in main experiments and Qwen3-14B (Yang et al. 2025a) in additional experiments as the base model. We utilize the VeRL (Sheng et al. 2025) framework with VLLM (Kwon et al. 2023) inference engine to distribute the training workload efficiently. The experiments are executed on a cluster of 8 NVIDIA A100-SXM4-80GB GPUs. During the trial steps of TTED and data collection of baseline methods (as described in Methodology section), the sampling temperature is set to 1.0, and thus obtain the rollout probabilities $\pi_{\theta_{old,inf}}$ directly from the inference engine. The training is performed on all parameters of the model. For RL, the maximum input token length is restricted to 16,384 tokens, and the maximum output token length is set to 4,096 for action generation and 6,144 for

environment decomposition; and for SFT, the whole trajectory is left-truncated to 22,528 tokens. The overall training time for all methods is shown in Table 10. In addition to total GPU hours, Table 10 also reports the test-time exploration costs, including average inference time, token usage, and interaction steps per sub-environment exploration. TTED incurs higher per-step exploration costs than TTT, but its total training cost remains comparable to TTT and lower than TTRL. Together with the performance improvement on WebArena (21.0% vs. 12.5%), these results show that TTED trades additional exploration costs for more effective adaptation rather than being cost-free. Table 6 further shows that TTED reduces prompt lengths for action generation. Training with ground-truth labels requires less time because the overall accepted training data is fewer.

Adv. Estimation	Method	Inf. Time (s)	#Tokens	#Steps	GPU h.
SFT	SFT w/ GT	—	—	—	2.24
	TTT	—	—	—	11.07
	TTED	—	—	—	12.53
REINFORCE	RL w/ GT	—	—	—	8.67
	TTT	16.7	17514.9	6.17	14.40
	TTED	46.8	26114.5	9.91	14.80
GRPO	TTRL	34.9	25808.0	4.54	19.20
	TTED	101.9	29231.7	5.76	21.33

Table 10: Computational costs for different adaptation methods on WebArena. All methods are evaluated under a uniform training budget of 3200 sub-environments. GPU hours are estimated with Qwen3-8B on 8 NVIDIA A100-SXM4-80GB GPUs. Avg. inference time and token numbers are measured per sub-environment exploration.

Data Collection and Trajectory Rollout For the trial steps defined in Methodology section, the agent interacts with the decomposed sub-environments of WebArena to gather trajectories, including accepted and rejected trajectories of action generation and environment decomposition. In addition, we filter out action generation trajectories with actions outside the action space with rules. For all methods, the training batch is constructed by sampling trajectories from 32 sub-environments, with one action generation and two environment decomposition trajectories, i.e. sub-goal and sub-observation decomposition (if applicable), per sub-environment. To ensure a fair comparison between SFT and RL methods, we obtain all data before training and reuse the same set of trajectories for both SFT and RL training. For methods with GRPO, we collect an additional set of trajectories by sampling multiple actions per sub-environment, resulting in the total group size of four, to compute the self-consistency reward. For the self-consistency matching, two actions are considered a match if they are identical in their action type and parameters, except (1) the “noop” and “report_infeasible” actions are ignored for parameter matching; and (2) the “fill” and “send_msg_to_user” actions are considered a match when their textual parameters are semantically similar, or concretely, over 0.75 cosine similarity of embeddings on all-MiniLM-L6-v2 (Reimers and Gurevych 2019; Wang et al. 2020). And the majority will be assigned rewards equal to one, and others are rewarded zero. The training pro-

cess is conducted for at maximum 100 steps, resulting in a total of 3200 sub-environments for training.

Model Training in TTED As detailed in Methodology section, we adopt an off-policy REINFORCE algorithm augmented with rollout importance sampling and advantage normalization from REINFORCE++ (Hu et al. 2025). The policy optimization employs the Group Sequence Policy Optimization (GSPO) loss (Zheng et al. 2025).

- **Reward Normalization:** Rewards are assessed pointwise and independently for each trajectory by using the backbone model as a GRM. The according prompt template, including principles and scoring pattern, is elaborated in Appendix E. The raw scores from the GRM—specifically $r_{\text{goal}}, r_{\text{obs}} \in [0, 5]$ and $r_{\text{action}} \in \{0, 1\}$ —are mapped to a bounded scalar return $r_t^* \in [-1, 1]$ with linear transformation.
- **Advantage Estimation:** For the default REINFORCE-based setting, we compute the normalized advantage $\hat{A}_t = \frac{r_t^* - \mu_{r^*}}{\sigma_{r^*} + \epsilon}$ across the training mini-batch to stabilize the test-time training process. For the SFT variant, we remain trajectories with rewards larger than 0 after normalization; for the GPRO variant, we use self-consistency rewards as described above, apply the original GRPO advantage estimation, and filter out groups whose actions receive all the same reward value.
- **Optimization Hyper-parameters:** The peak learning rate for RL is set to 10^{-6} with none-warmup schedule and a 0.1 decay; and the learning rate for SFT is 10^{-5} . These settings are resulted from a grid search on $\{5 \times 10^{-7}, 10^{-6}, 2 \times 10^{-6}, 5 \times 10^{-6}, 10^{-5}\}$. To prevent model degradation and adaptation drift, we apply a gradient clipping with a threshold of 5.0 and, for RL, IS ratio clipping with an upperbound of 0.01 and a lowerbound of 0.005. We do not use KL divergence regularization, as we find it does not contribute to the performance improvement in our preliminary experiments.

Since it is not allowed for us to test the checkpoints on WebArena in the middle of test-time training, we also split an 0.1 ratio validation set, and observe the GSPO loss value to stably descend on the validation set as the training goes. So that, the final checkpoint is selected as the last checkpoint.

Model Training in Baseline Methods For the “SFT w/ GT” and test-time SFT (“TTT (SFT)”) baselines evaluated in Table 1, the models are fine-tuned purely on the successful demonstration trajectories decided by the ground truth labels or agent self-assessed rewards (larger than 0 after normalization), respectively. For ground truth labeling, if a multi-turn trajectory is labelled success, we treat its every single turn as a success for training. Even though, the training set of “SFT w/ GT” is too small that we train the model with two epoches (still within the maximum of 100 training steps). For test-time RL (“TTT (RL)”) and TTRL, we also use GSPO loss with REINFORCE++-based and GRPO-based advantage estimation approaches, respectively. Other settings are the same to the corresponding variants of TTED.

D.3 Implementation Details of TTED

In this section, we provide detailed settings for TTED, including hyperparameters and model configurations for each dataset evaluation.

Evaluation on CompWoB+. We evaluate TTED with in-context learning (ICL) on CompWoB+ (Furuta et al. 2024). We describe the experimental setup and hyperparameter settings in the following.

Base Model: For evaluation on CompWoB+, we use GPT-4o-mini-2024-07-18 (Hurst et al. 2024) and Qwen3-8B (Yang et al. 2025a) as the base model. We access GPT-4o-mini-2024-07-18 via the API. For Qwen3-8B, we deploy the model using VLLM (Kwon et al. 2023) and run inference on 4 NVIDIA A100-SXM4-80GB GPUs.

Pipeline: Since CompWoB+ consists of single-turn static webpages, we adopt TTED with in-context learning (ICL) for evaluation. Unlike real-world web environments, the entire environment in CompWoB+ can be decomposed at once into multiple sub-environments. The agent first attempts to solve tasks within these sub-environments, and the resulting trajectories are used as experience to guide final action generation in the original environment via ICL. The overall workflow is illustrated in Figure 10. In particular, for the self-assessment process in the sub-environment, we further improve reliability through simulation. Specifically, we first construct a simulated environment using Selenium (Selenium 2026) based on each sub-environment. The generated actions are then executed in the simulated environment to obtain execution feedback. In addition, we employ the LLM to generate JavaScript code to verify whether the actions are correctly executed. Finally, the execution feedback, JavaScript-based verification signals, and generated actions are provided to evaluate whether the action in the current sub-environment is correct. If the actions is judged as successful, the attempt in the sub-environment is recorded as experience and used to guide the final action generation in the original environment. Otherwise, the agent retries within the sub-environment. See Figure 11 for detail.

Hyper-parameters: We introduce several hyperparameters to balance reliable pipeline execution and efficiency. When errors occur during execution, we employ retry mechanisms to improve robustness. Specifically, if the LLM output does not satisfy the required format and answer extraction fails, the model retries up to 5 times. Additionally, when the self-assessment module determines that the actions fails within the sub-environment, the agent re-attempts completion within the same sub-environment for up to 3 attempts. If the actions still fails after three attempts, the current sub-environment is discarded. For each interaction with the LLM, we limit the maximum input length to 30,000 tokens, output length to 5,000 tokens. For Qwen3-series models, we further allocate an additional thinking budget of 5,000 tokens.

Rollouts on WebArena. To demonstrate the ability to tackle realistic web automation tasks, we adopt TTED with RL to improve performance on WebArena. The rollout strategy and settings are described as follows.

Base Model: We use Qwen3-8B (Yang et al. 2025a) as the base model for TTED in the WebArena sampling. We deploy Qwen3-8B using the VLLM (Kwon et al. 2023) inference framework and run inference on 4 NVIDIA A100-SXM4-80GB GPUs.

Pipeline: The rollout pipeline is shown in Figure 3. At each step, the agent performs environment decomposition based on the current environment, producing a sub-goal and a corresponding sub-observation. Conditioned on the decomposed sub-environment, the agent then generates an action for the current step. Before executing this action in the original environment, the self-assessment module evaluates both the quality of the decomposition and the correctness of the generated action, including whether the sub-goal and sub-observation are appropriate for the current state. Only when both the decomposition and the generated action pass self-assessment is the action executed in the original environment to advance the task. Otherwise, the agent re-performs environment decomposition on the current environment and starts a new attempt. The summary module then records the action information as part of the interaction history.

Through this process, we collect a large number of sub-environments along with their corresponding self-assessment signals on WebArena. These sampled trajectories are then used as training data for subsequent RL optimization.

Hyper-parameters: During rollout on WebArena, we set the maximum number of environment steps to 15. If the step limit is exceeded, the task is terminated. When the LLM output fails to satisfy the required format, we retry generation for up to five times. If the output still fails after 5 attempts, the task is considered failed and terminated. We set the sampling temperature to 1.0. In the self-assessment stage, the action is evaluated as either pass or reject, while the sub-goal and sub-observation are scored on a scale from 0 to 5. We provide a detailed description of the evaluation principle for actions, sub-goals, and sub-observations in Appendix E. A sub-environment is considered successful only when the action is evaluated as pass and both the sub-goal and sub-observation receive scores of at least 3. For each LLM call, the maximum input length is limited to 30,000 tokens, and the maximum output length is limited to 5,000 tokens. In addition, we allocate a reasoning token budget of 5,000 tokens. If any of these limits are exceeded, the corresponding inputs or outputs are truncated.

Evaluation on WebArena. We evaluate TTED with RL on WebArena (Zhou et al. 2024), a real-world web environment benchmark. The experimental setup and hyperparameter settings are described as follows.

Base Model: We use Qwen3-8B (Yang et al. 2025a) and our trained versions as the base model for TTED in the WebArena evaluation. We deploy Qwen3-8B using the VLLM (Kwon et al. 2023) inference framework and run inference on 4 NVIDIA A100-SXM4-80GB GPUs.

Pipeline: During evaluation, we remove the self-assessment module, while keeping the rest of the pipeline consistent with the sampling stage. Specifically, the agent first performs environment decomposition to obtain sub-environments, then generates actions within each sub-environment, and finally records the actions through the summary module. This process is repeated iteratively until the task is completed or reach the maximum steps.

Hyper-parameters: During evaluation, the maximum number of environment steps is set to 15. For the LLM, we set the temperature to 0. All other settings follow those used in the rollout stage, including a maximum input length of 30,000 tokens, a maximum output length of 5,000 tokens, and a reasoning token limit of 5,000. When the LLM output fails to satisfy the required format, we retry generation for up to 5 attempts.

E Prompt Templates

In this section, we provide the LLM prompts used in the implementation of TTED, including the TTED with in-context learning (ICL) method applied to single-page web tasks in CompWoB+ (Furuta et al. 2024), and the TTED with RL method used in realistic web environments WebArena (Zhou et al. 2024).

E.1 Prompt for Single-Page Web Tasks

We perform environment decomposition through two stages: Goal Decomposition and Observation Decomposition. First, we decompose the overall task goal into multiple sub-goals. In our implementation, we provide the names of the base tasks that constitute the CompWoB+ dataset, along with the current task name. Based on this information, the model extracts the corresponding sub-goals from the overall task objective.

Next, in the Observation Decomposition stage, the model extracts the corresponding sub-observations based on the identified sub-goals and the full observation. Each pair of sub-goal and sub-observation forms a sub-environment.

Goal Decomposition

Given the composite task name and its goal description, decompose the composite goal into exactly some sub-goals. For each sub-goal, output the sub-goal name, and a corresponding concise natural language instruction. There are ten types of sub-goals: ‘login-user’, ‘click-link’, ‘click-widget’, ‘click-button’, ‘click-option’, ‘enter-time’, ‘choose-list’, ‘click-checkboxes’, ‘edit-text’, ‘read-table’. Note: A sub-goal does not imply a particular operation or goal, one sub-goal may involve multiple operations, for example, ‘click submit button’

is often combined with other operations to form a single sub-goal.

Current Task

Task Name: {task_name}

Goal: {goal}

Decomposed Sub-goals:

Observation Decomposition

Instruction

Given the basic information of a compositional goal, the corresponding original web observation (HTML), and a list of its decomposed sub-goals, your work is to decompose the original complex environment into a series of simpler sub-environments, each includes a specific sub-goal and a corresponding sub-observation. For each sub-observation:

Requirement 1: Preserve DOM Consistency with the Original Task

Each sub-observation must retain the same DOM structure and attributes (e.g., id, class, name, type, value, placeholder, etc.) of the target element as in the original HTML. This ensures that the same Selenium code (XPath, CSS selectors, etc.) used in the original observation will work identically in the decomposed observations.

Requirement 2: Include Required Context and Functional Components

In addition to the target element, each sub-observation must include any necessary surrounding or functional components required for the element to be visible, interpretable, and operable. These may include: <form> tags and their attributes like action, method, Related labels, container <div>s, submit buttons, Necessary parent elements, classes, or structure used in the original HTML for styling or scripting purposes.

Requirement 3: Ensure Standalone Executability

Each sub-observation should be a self-contained and standalone HTML file with a valid document structure (<!DOCTYPE html>, <html>, <head>, <body>, etc.). It must be directly openable in a browser and fully functional for manual interaction and Selenium-based automation without relying on external scripts or styles from the original page.

Requirement 4: Ensure Output Confirms to the Required Format

The name of each decomposed sub-goal in your answer should be wrapped in <subgoal> sub-goal content </subgoal>, and HTML content should be wrapped in <subobs> html content </subobs>.

Example:

<subgoal> subgoal 1: ... </subgoal>

<subobs>

HTML Content

</subobs>

Current tasks:

Task Name: {task_name}

Goal Description: {goal}

Decomposed Sub-goals: {sub_goals}

Original HTML: {ori_html}

Decomposed HTML:

Trial Action Generation

You are an autonomous computer control agent that can perform atomic instructions specified by natural language to control computers. There are two types of instructions it can execute.

{action_space_description}

Current Task

Below is the HTML code of the webpage where the agent should solve a goal.

Current observation: {sub_obs}

Current goal: {sub_goal}

Directly output the instructions, and do not include any thought and explanatory comments in your response.

Answer:

For the Script-Based Simulation component, we first predict the expected outcome of the generated action, and then generate corresponding JavaScript code to verify the predicted effect. Additionally, we explicitly inform the model that, in CompWoB, the submit button does not behave as a typical submission action and does not trigger page navigation.

Script-Based Simulation

You are given a HTML snippet, a goal and a current user action that is executed on this HTML using Selenium.

Your tasks are:

1. Analyze what change or effect the action is expected to cause in the web page.

Attention! All buttons named “submit” or containing “submit” do not have actual submission functionality; clicking them does not trigger any effect. Therefore, to verify this action, it is sufficient to check whether the button is being clicked.

2. Generate JavaScript code that can be executed to verify whether the action was correctly performed. The generated JavaScript code must include a return value, and the return value must be either true or false. Do not wrap the generated JavaScript code in a function. Note! The generated JS code must comply with JavaScript syntax standards and ensure that the code can be executed properly.

Example JavaScript code

```
var      selectElement      =      docu-
ment.getElementById(“options”);
```

```
return selectElement.value === “Cameroon”;
```

3. Return the result in the following format:

Response Format:

Analysis:

<text>

<Brief explanation of the expected change on the page>

</text>

Verify Javascript code:

```

<code>
<JavaScript code that checks whether the action was
successful>
</code>
# Example
Action: clickxpath //select[@class='generated-select'
and @id='options']
Analysis:
<text>
Observation: The action is expected to trigger the drop-
down selection process, allowing the user to choose
one of the options displayed in the '<select>' element.
However, since the action performed is just clicking on
the '<select>' element, it might not directly result in a
change until a specific option is selected. The verifica-
tion can check if any interaction was triggered success-
fully.
</text>
Verify Javascript code:
<code>
var      selectElement      =      docu-
ment.getElementById("options");
return selectElement === document.activeElement; //
Verifies if the select element is focused after the click
</code>
# Current task:
Goal: {sub_goal}
HTML: {sub_obs}
Action: {action}
JS code:

```

For the Self-Assessment stage, the model evaluates whether the generated action is successful by considering the observation, the sub-environment, and the execution results obtained from the simulated environment. Based on this information, the model provides a final judgment of either pass or reject.

Self-Assessment

You are a helpful assistant. What you need to do is determine whether the instruction can be successfully executed on the web page and whether it achieves the corresponding goal.

```

{action_space_description}
Current HTML Page: {sub_obs}
Current Goal: {sub_goal}
Current Instructions: {actions}

```

Every instructions is executed on the web page and corresponding feedback can be obtained. Specifically, for each command, the following information is provided:

Action: The action that was executed

Observation: The change observed on the web page after the action

Inspect result: Whether the action was executed correctly

Here are the instructions' information:

```

{js_simulation}
Your Job:

```

1. Analyze: Determine whether the given instructions can successfully accomplish the goal on the provided HTML page. Please analyze each instruction one by one to determine whether an error occurred.
 2. These instructions are executed within the web page; please make judgments based on the feedback from their execution.
 3. Attention! All buttons named "login", "reset", "submit" or containing "submit" do not have actual submission functionality; clicking them does not trigger any effect. Similarly, descriptions in the goal such as click "submit", "login" and "reset" only require clicking the corresponding buttons, without the need for actual submission functionality.
 4. Give your judge result. Attention output the final answer enclosed in \$\$.
- If they are correct, respond with: \$\$pass\$\$.
If they are incorrect or incomplete, respond with: \$\$reject\$\$.
5. Based on the analysis, provide the recommend answer to the instruction. If the evaluation result is "pass" then the recommend answer is the original answer; if the result is "fail" generate a new recommend answer based on the analysis.
- Response Format:
- Analyze:
1. ...# Instruction 1 analysis
 2. ...# Instruction 2 analysis
 - ...
- Judge result: \$\$pass\$\$ or \$\$reject\$\$
- Recommend answer:...

Final Action Generation

You are an autonomous computer control agent that can perform atomic instructions specified by natural language to control computers.

```

{action_space_description}

```

Here are some examples for specific tasks, please follow the example to generate your answer for the current task.

```

{sub_environments}

```

Below is the HTML code of the webpage where the agent should solve a goal.

```

# Current task
Current html: {observation}
Current goal: {goal}
Answer:

```

E.2 Prompt for Realistic Web Automation Tasks

For realistic web automation tasks, we also perform goal decomposition followed by observation decomposition. Unlike single-page web tasks, it is not feasible to generate all sub-environments at once. Instead, the process is performed iteratively. At each step, the agent generates the next sub-goal based on the overall objective, the current observation, and the historical trajectory. The observation decomposition module then extracts the relevant sub-observation according to the generated sub-goal and the current observation, thereby forming a sub-environment.

Goal Decomposition

You are a web agent planner.

Your job is to analyze the current web environment and the agent's progress so far, and then determine the most appropriate next sub-task the agent should perform.

You are given the following information:

1. **Task Goal**: The ultimate objective the user wants to achieve.
2. **Current Page (AXTree)**: A structured textual representation of the current page's accessible elements.
3. **Action History**: The list of actions the agent has performed so far in order.
4. **Failure Attempts**: A list of previously attempted actions that failed on this page, along with the reasons for their failure.

Your job:

- Reflect on the task goal, the current state of the page, and what has already been tried.
- Identify the most reasonable and necessary next sub-task to move the agent closer to the goal.
- The sub-task should be high-level but specific, and only describe the next immediate step.
- Do not provide low-level commands such as `click('277')`. Instead, describe the sub-task using high-level natural language — for example: "Click the link labeled 'The A11Y Project / a11yproject.com' to open the project page."
- The sub-task should be as simple as possible, each containing only one objective and requiring only a single step to complete.

Output format:

Next Sub-task:

[One concise, natural language instruction describing the agent's next step.]

Current task

Task Goal: {goal}

Current Page (AXTree):

{axtree}

Action History:

{history}

Failure Attempts:

{history.get_error_descendants()}

Observation Decomposition

You are an intelligent web agent. Your task is to extract the goal-relevant sub-observation from a full **Accessibility Tree (AX Tree)** of a webpage.

Given:

- A user goal description.
- The full AX Tree text representation of the current web page.

Instructions:

- Select all AX Tree fragments (i.e., nodes and their subtrees) that are directly or indirectly relevant to the goal or sub-goal.

- Err on the side of inclusion: **if you are unsure whether an element is needed, include it.**

- You must **copy the selected fragments exactly as-is** from the original AX Tree.

- **Do not modify** any content — including the AXNode ID, node type, labels, values, or attributes.

- AXNode IDs (e.g., '[163]') must **match the original AX Tree exactly**.

- You may extract multiple non-contiguous branches if needed.

- Capture as much of the environment as possible to avoid omissions or errors.

Output format:

[The extracted sub-observation]

Current task: Goal: {sub_goal}

Current Page

Accessibility Tree:

{axtree}

Action Generation

You are a web assistant. You will be given web-based tasks. These tasks will be accomplished through the use of specific actions you can issue.

Here's the information you'll have:

The user's objective: This is the final goal you're trying to achieve.

The sub-goal: This is the current sub-goal you need to perform in order to move toward the final goal.

The current web page's accessibility tree: This is a simplified representation of the webpage, providing key information.

The current web page's URL: This is the page you're currently navigating.

The open tabs: These are the tabs you have open.

The history actions: These are the history actions you have done, including actions and its intent.

The failure attempts: These are failed attempts made on the current page, including a failure summary, cause analysis, and suggestions. Please avoid making similar mistakes when performing actions.

You can interact with the environment using the following actions:

{action_space_description}

To be successful, it is very important to follow the following rules:

1. You should only issue an action that is valid given the current observation

2. You should only issue one action at a time.

Your output should be included:

- Clearly explain what the action is trying to achieve (intent)

- Clearly state which element is being operated on (element ID + label/type)

- Output the final action command that should be executed

Output format:

```

### Manipulate element:
[Identify the specific element that the agent will interact
with.]
### Action:
[Write the exact action command that should be executed
on the page.]
### Action intent:
[Summarize what the action is trying to accomplish in
the context of the task.]
Here is an example of how to use the bid action:
click('314')
Here is an example:
# Example
{simple_example}
# Current Task
Observation:
{sub_obs}
URL: {obs["url"]}
Final goal: {obs["goal"]}
Sub-goal: {sub_goal}
History Actions:
{history}
Failure attempts:
{history.get_error_descendants()}
Action:

```

In the self-assessment stage, we evaluate whether the generated action and the corresponding sub-environment are correct. To define correctness, we specify a set of principles that guide model for accurate evaluation. For actions, the generated action must be executable and capable of achieving the sub-goal. The action is evaluated in a binary manner, with outcomes labeled as either pass or reject.

For the sub-goal and sub-observation, we design a scoring scheme ranging from 0 to 5. Specifically, we expect the sub-goal to be closely related to the overall objective, achievable with a single action (i.e., atomic), clearly defined without ambiguity, and free from hallucinations. For the sub-observation, we require that the elements remain consistent with the original observation, include all elements necessary to accomplish the sub-goal, and preserve the structural organization of the original observation. Based on these criteria, the final scores are assigned.

Self-Assessment

You are an expert web automation evaluator, responsible for assessing both action execution success and the quality of goal/observation decomposition made by an AI agent during a web task.

Your judge combines two perspectives:

Action Execution Accuracy — Did the executed action actually achieve its intended purpose?

Decomposition Quality — Was the sub-goal (and its supporting sub-observation) well-defined, logically placed, and consistent with the overall task?

You will be provided with:

1. Overall Goal — the agent's high-level objective.
2. Overall Observation of Current Step — the full ob-

servation before sub-observation extraction.

3. Action History — a list of previously executed steps.

4. Executed Action — the specific action the agent just performed.

5. Detailed Element — the DOM element involved in the action.

6. Action Intent — the natural-language description of what the agent wanted to achieve with the action.

7. Sub-goal — the decomposed step corresponding to this action.

8. Sub-observation — the extracted sub-observation supporting that sub-goal.

Part 1: Action Execution Success

You must determine whether the executed action successfully fulfilled its intent.

- Based on the action and observation, determine whether the action can complete the sub-goal.

- If the action did not align with sub-goal exactly but still contributes positively toward achieving the overall goal, count it as successful.

- Provide a brief explanation of why the action succeeded or failed.

Output either:

true or false inside code fences for the result.

Part 2: Goal and Observation Decomposition Quality

Sub-goal Quality (0-5)

A high-quality sub-task should be atomic, precise, and logically aligned with both the task goal and previous steps.

Scoring reference:

```
{goal_decomposition_principle}
```

Here are some examples for goal quality judging:

```
{sub_goal_judge_example}
```

Sub-observation Quality (0-5)

A valid sub-observation must be a verbatim subset of the original observation and contain all elements necessary to perform the sub-goal.

Scoring reference:

```
{obs_decomposition_principle}
```

Required Output Format (JSON)

```
{
```

```
  "action_success": true or false,
```

```
  "action_reasoning": "why the action succeeded or failed",
```

```
  "subgoal_score": int (0-5),
```

```
  "subobs_score": int (0-5),
```

```
  "subgoal_reasoning": "1-2 sentences explaining the sub-task quality judge",
```

```
  "subobs_reasoning": "1-2 sentences explaining the sub-environment quality judge",
```

```
}
```

Current Evaluate

Overall Task Goal: {goal}

Overall Environment of Current Step: {obs}

Previous Steps: {history}

Current Step Sub-task: {subgoal}

Current Step Sub-environment: {subobs}

Executed Action: {action}

Detailed Element: {element}
Action Intent: {action_intent}

Self-Assessment Principles

Goal Decomposition Principle:

- 5: Clear, atomic, non-redundant, consistent with the goal.
- 4: Slightly vague but mostly correct.
- 3: Somewhat unclear or contains mild future assumptions.
- 2: Poorly scoped or confusing.
- 1: Directionally wrong or ambiguous.
- 0: Not actionable or irrelevant.

Observation Decomposition Principle:

- 5: Fully correct, complete, and structurally consistent.
- 4: Minor omissions but still workable.
- 3: Noticeable omissions yet partially usable.
- 2: Missing key elements, difficult to use.
- 1: Nearly unusable.
- 0: Contains fabricated or altered elements.

Summary

You are a Summarization Agent in a web automation system.

Your role is to produce a concise, accurate, and context-aware summary of the current step.

Your summary must reflect (1) the current page environment, (2) what element was manipulated, (3) the action's intent, and also incorporate the judge result to reflect whether the step was meaningful or successful.

You are given:

- 1. **Task Goal**: The overall objective of the task.
- 2. **Overall observation of Current Step**: The full environment of current step.
- 3. **Sub-goal**: The current sub-goal the agent was performing.
- 4. **Action**: The action will be executed.
- 5. **Manipulated Element**: The element that is manipulated by the action.
- 6. **Action Intent**: A description of what the action is trying to achieve.
- 7. **Judge Result**: The evaluation result indicating whether the action succeeded in achieving its intent and the quality of the sub-task and sub-environment.

Your summary must include:

- 1. **Page Observation Description**: Briefly describe the functional nature of the current page, such as:
 - "an online shopping page with a product search bar"
 - "a login page requiring username and password input"Do NOT list specific DOM nodes. Summarize the observation as a key functional areas relevant to the action.
 - 2. **Manipulated Element**: Describe which element was interacted with, in natural language.
 - 3. **The Summary of the Action**: Explain what the action achieve based on the given information.
- Do not include reasoning or explanation — just the sum-

mary.

Required Output Format (JSON)

```
{
  "observation_description": "a brief description of the
functional areas of the page",
  "manipulated_element": "the element that is manipu-
lated by the action",
  "action_summary": "a description of what the action
achieve",
}
```

Current task
Task Goal: {goal}
Overall observation of Current Step: {axtree}
Sub-goal: {subgoal}
Action: {action}
Manipulated Element: {element}
Action Intent: {intent}
Judge Result: {judge}

F AI Tool Usage Disclaimer

AI-assisted tools are used solely for grammar correction, language polishing, and minor wording improvements in this manuscript. Their usage is strictly limited to linguistic and editorial assistance. All core ideas, methodological contributions, and experimental designs are developed entirely by the human authors.